Communicative Efficiency in Adjective Ordering: Plan-Guided Production between Hierarchical Composition and Incremental Pragmatics

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Abstract

Communicative efficiency has been invoked to explain both stable adjective-order preferences and their sensitivity to referential context. However, these accounts differ in how adjective order supports efficient communication. Under hierarchical composition, a reliable modifier near the noun can help the listener interpret a more subjective adjective. During incremental production, a discriminating modifier early in the utterance can help the listener identify the referent sooner. These benefits can favour opposing orders, raising the question of how speakers reconcile them when selecting and ordering adjectives. We propose a plan-guided incremental speaker model that jointly predicts adjective selection, utterance length, and order by combining utterance-level utility with competition during incremental adjective choice. Bayesian comparison of theory-informed probabilistic pragmatic models fitted to German production data favoured this account over global and fully incremental alternatives. Within this framework, context-fixed semantics predicted responses more accurately than sequential context updating, chiefly through better predictions of adjective selection and length. Across two German slider studies, size-first preference was strongest when size alone identified the target and weakest when colour alone did, with a small size-first preference remaining in the latter context. In production, speakers often selected a single sufficient adjective; when both size and colour were included, size-first order remained more frequent. Lower size discriminability increased redundant adjective use most clearly when size was sufficient. Controlled simulations showed that allowing single-adjective alternatives changed the average listener advantage from colour-first to size-first under both semantic regimes: the availability of shorter descriptions changed how the listener interpreted the two-adjective utterances. Together, these findings support a unified account in which stable ordering preferences and contextual flexibility interact through the selection of a description's content, length, and adjective order.

Keywords: adjective ordering, reference production, Rational Speech Act, incremental production, Bayesian cognitive modelling

1 Introduction

Adjective ordering provides a window onto how stable order preferences interact with scene-specific informativeness in referential communication. In languages with prenominal modification, robust ordering preferences favour size-first sequences such as *the big blue sticker* over *the blue big sticker* (Sproat and Shih, 1991; Scontras, 2023). Yet the first adjective is also the listener's first descriptive cue to the target. In a scene where colour alone identifies the intended sticker, *blue* would provide the decisive information sooner, even though a colour-first sequence conflicts with the canonical size-first preference (Fukumura, 2018; Rubio-Fernández, Mollica, and Jara-Ettinger, 2021). The order that sounds most natural may therefore differ from the order that guides the listener most efficiently through the scene.

But what counts as efficient—and therefore early—for the listener in a complex referential setting? One relevant line of work concerns adjective subjectivity. More subjective adjectives tend to precede less subjective adjectives in prenominal sequences (Scontras, Degen, and Goodman, 2017). Some subjectivity-based proposals connect this tendency to hierarchical semantic composition, under which the noun-adjacent adjective composes before adjectives farther from the noun. In *the big blue sticker*, the noun-adjacent adjective *blue* therefore combines with *sticker* before the linearly earlier *big*, narrowing the comparison class against which the gradable size adjective is interpreted. Because *blue* is typically less subjective than *big*, its noun-adjacent position may provide a comparatively stable restriction before the size adjective is interpreted, potentially contributing to the surface size-first preference (Franke, Scontras, and Simonic, 2019; Scontras, Degen, and Goodman, 2019; Scontras, Bar-Sever, et al., 2020). A separate line of referential work suggests that adjective order is also sensitive to the discriminatory strength of each property in the current scene (Fukumura, 2018; Rubio-Fernández, Mollica, and Jara-Ettinger, 2021). When colour distinguishes the target more effectively than size, colour-first orders may become more acceptable or more likely, consistent with a pressure to provide contextually useful information earlier in the surface sequence.

Depending on which perspective is taken, communicative efficiency may support conflicting adjective-ordering predictions. This tension is especially consequential in reference games (Franke, 2009; Frank and Goodman, 2012), where successful communication depends both on the completed description and on how information becomes available as that description unfolds. Work on incremental language processing emphasizes the latter: listeners integrate partial linguistic input with visual context to restrict candidate referents before the noun phrase is complete (Tanenhaus et al., 1995; Sedivy et al., 1999). On this view, placing the most discriminating modifier early in the linear sequence may facilitate identification before the remaining words are heard (Rubio-Fernandez and Jara-Ettinger, 2020). On the other hand, the subjectivity-based account emphasizes the former by evaluating a completed adjective sequence through the listener's hierarchical interpretation. From this comprehension-oriented perspective, an order is efficient when the noun-adjacent adjective provides a more reliable restriction for interpreting the outer adjective (Franke, Scontras, and Simonic, 2019; Scontras, Degen, and Goodman, 2019; Scontras, Bar-Sever, et al., 2020).

Moreover, the ordering choices available to a speaker depend on which properties are selected and how much information the description includes. In a colour-sufficient scene, *the blue sticker* may complete the utterance, so competition between size–colour and colour–size orders may never arise. In a size-sufficient scene, speakers may still produce *the big blue sticker*, including colour even when size alone identifies the target. This creates an asymmetry in the redundant use of colour and size adjectives. Previous empirical studies show that colour is included redundantly more often than relative adjectives such as size (Tarenskeen, Broersma, and Geurts, 2015; Rubio-Fernández, 2016). Such redundancy can still support efficient identification, especially when modifier meanings are uncertain and gradable, as with size adjectives (Rubio-Fernández, 2016; Degen et al., 2020). Scene-specific usefulness may therefore affect which adjectives are selected and when production stops, as well as how jointly selected adjectives are ordered.

These choices make the full distribution of one-, two-, and three-adjective utterances the relevant production outcome. Within this response space, adjective order is conditional on adjective selection and utterance length. The usefulness of each adjective also depends on the other properties included in the description. This dependence raises a question about production architecture. A global speaker compares complete utterances as wholes. A fully incremental speaker evaluates each next adjective against the alternatives available after the words already produced. Between these endpoints, an utterance-level plan may constrain production while local alternatives continue to influence incremental word choice. We call this intermediate hypothesis *plan-guided incremental production*. Under this hypothesis, an utterance-level evaluation combines with the scene-specific usefulness of incremental adjective choices. This possibility connects to broader literature on bounded planning, under which speakers can begin speaking before every part of an utterance has been prepared and can continue planning during articulation (Ferreira and Swets, 2002; Konopka and Meyer, 2014).

Our central research question is: *How do speakers reconcile the pressures from hierarchical semantic composition and incremental production when selecting and ordering adjectives?* We address this question through three more specific questions, moving from the empirical patterns to computational accounts of the mechanisms that may generate them. First, how do stable size-first preferences and scene-specific referential usefulness jointly shape explicit order judgements and the production outcomes of adjective selection, utterance length, and conditional order? Second, which production architecture best predicts this distribution: global evaluation of complete utterances, fully incremental adjective choice, or plan-guided incremental production between these endpoints? Third, does updating the context for interpreting size improve prediction of the production responses?

To address these questions, we combine controlled behavioural experiments, computational cognitive modelling, simulations across a broader range of randomly generated contexts, and Bayesian predictive model comparison. The empirical programme separates explicit ordering preferences from the production choices that make an order observable. Studies 1a and 1b use a slider-rating task, with Study 1b providing a direct replication in a new sample. Both measure participants' judgements of size-first and colour-first referential utterances across referential contexts. Study 2 uses direct production to elicit utterances and records adjective selection, utterance length, and order conditional on producing the same adjective set. By varying referential context and size discriminability, we separate the relative usefulness of size in a particular scene from its perceptual reliability throughout the task. This manipulation tests the role of perceptual reliability while holding lexical subjectivity constant.

On the modelling side, we formalise these questions within the Rational Speech Act framework (Frank and Goodman, 2012; Goodman and Frank, 2016; Degen, 2023). We extend the qualitative model of Schlotterbeck and Wang (2023) from two fixed adjective orders to the full production distribution. That account combined sequential context updating within hierarchical semantic composition with incremental speaker choice. The separate predictive contributions of these two mechanisms remained unresolved. All production models predict the same distribution of one-, two-, and three-adjective responses involving size, colour, and form. The inclusion of form extends the comparison to utterances in which several properties can compete for mention and initial position. The models share controls for stable order expectations, visual salience, adjective selection, and utterance length.

Predictive model comparison asks which components improve prediction of the observed production distribution (Vehtari, Gelman, and Gabry, 2017). A separate simulation compares the semantic regimes on the same randomly generated contexts to ask when updating generates an ordering asymmetry and whether that asymmetry also arises under context-fixed semantics. By introducing the production model's utterance evaluation and broader alternatives in controlled steps, it examines how the communicative advantage changes when ordering is considered together with adjective selection and utterance length.

Empirically, the slider studies reveal a replicated context gradient: size-first preferences are strongest when size identifies the target and weakest when colour does. A small residual size-first preference remains when colour alone identifies the target. The production study shows that speakers often resolve reference with one adjective, so adjective selection and stopping frequently determine whether an order contrast arises. When size differences are less distinct, speakers include more adjectives than are needed to identify the target, especially when size alone is sufficient.

After accounting for variation across participants in stable order preferences, the best predictions combine evaluation of complete descriptions with continued influence from incremental adjective choices. This pattern is consistent with plan-guided incremental production. Models that interpret size relative to the original scene predict production more accurately than models that update this context as adjective meanings combine. This predictive difference is concentrated in adjective selection and utterance length. In the simulations, allowing single-adjective alternatives changes the average listener advantage from colour-first to size-first under both fixed and updating interpretations. The communicative value of the two orders therefore depends on the other descriptions available to the speaker.

Section 2 develops the theoretical background, followed by the behavioural studies in Section 3. Sections 4 and 5 present the formal models and their predictive comparison; Section 6 examines the conditions for a communicative ordering advantage. Section 7 discusses the findings in relation to reference production and semantic composition.

2 Sources of Adjective Ordering Preferences

Research on adjective ordering has identified several stable predictors of preferred order, including meaning-based factors such as closeness to the noun and inherentness (Sweet, 1898; Whorf, 1945; Martin, 1969), class-based hierarchies (Quirk et al., 1972; Dixon, 1982; Cinque, 1994; Scott, 2002), and information-theoretic measures (Hahn et al., 2018; Futrell, Dyer, and Scontras, 2020; Dyer et al., 2023). These predictors capture stable regularities in adjective order, while leaving room for contextual variation (for overviews, see Dyer et al., 2023; Scontras, 2023). Referential context contributes further scene-specific variation in ordering preferences (Fukumura, 2018; Trotzke and Wittenberg, 2019). The present account brings together three lines of work: subjectivity and hierarchical semantic composition, discriminatory strength in reference production, and incremental pragmatic models. Their connection lies in how the usefulness of an adjective depends both on its interpretation and on its contribution to a developing description.

2.1 Subjectivity, Hierarchical Composition, and Adjective Order

An empirical generalisation relates an adjective's *subjectivity* to its preferred distance from the noun: more subjective adjectives are preferred farther from the noun (Scontras, Degen, and Goodman, 2017). Subsequent work suggests that this relation recurs across languages (Trainin and Shetreet, 2021; Scontras, 2023). The greater subjectivity of size adjectives relative to colour adjectives predicts a size-first preference.

A communicative explanation of this pattern rests on restrictive composition (Scontras, Degen, and Goodman, 2019). Subjective meanings allow greater divergence between speakers' and listeners' interpretations, which can make reference less successful. Under hierarchical semantic composition, the adjective closest to the noun restricts the context in which an outer adjective is interpreted. Applying a less subjective adjective first can therefore limit the domain over which a more subjective interpretation may diverge. Simulations formalise this proposed benefit and show that sequential restriction can increase the probability of identifying the intended target (Franke, Scontras, and Simonic, 2019). For gradable size adjectives, restriction can also change the comparison class against which size is evaluated. This mechanism of *incremental semantic restriction* connects subjectivity-based ordering to the communicative consequences of hierarchical composition (Scontras, Bar-Sever, et al., 2020). The

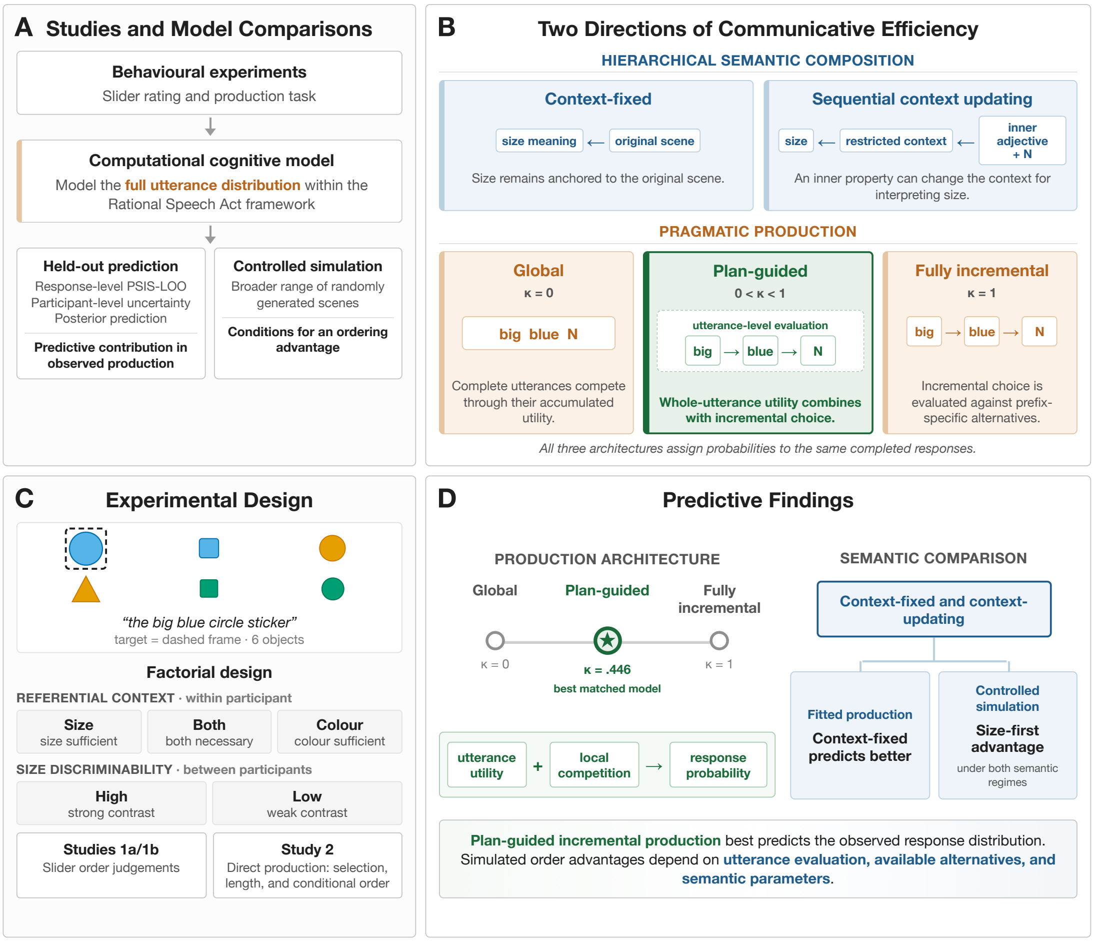


Figure 1: Overview of the design, model space, and evidential structure. Panel A traces the behavioural evidence through computational modelling to held-out prediction and controlled simulation. Panel B distinguishes sequential context updating within hierarchical semantic composition from global, plan-guided, and fully incremental production. Panel C shows the experimental design, and Panel D summarises the predictive result for plan-guided production and the fitted and simulated evidence for sequential context updating.

ordering pressure arises from the interpretation of the combined description and can therefore operate when a speaker evaluates complete utterances as wholes.

This account motivates examining whether context updating contributes to the descriptions speakers produce in a particular scene. For size, the relevant comparison is between an interpretation anchored to the original scene and one recomputed as another adjective restricts the context. We refer to these possibilities as *context-fixed* and *context-updating* semantics. Comparing them allows the contribution of semantic updating to be evaluated alongside the stable size-first preference.

2.2 Referential Usefulness, Adjective Selection, and Length

A second ordering pressure comes from referential usefulness in the surface linear sequence. Speakers are more likely to place the more discriminating adjective first, linking ordering to the relative usefulness of properties in the current scene (Fukumura, 2018). The *discriminatory strength* of a property is its ability to distinguish the target from the other objects in the scene. The same adjective can therefore be highly useful in one array and much less useful in another. Under the *incremental communicative efficiency hypothesis*, an early modifier is useful insofar as it narrows the listener's candidate set before the utterance is complete (Rubio-Fernandez and Jara-Ettinger, 2020; Rubio-Fernández, Mollica, and Jara-Ettinger, 2021). This proposal accords with evidence that listeners integrate partial linguistic input with visual context (Tanenhaus et al., 1995; Sedivy et al., 1999). The communicative benefit thus depends on when information becomes available in the surface sequence. This connects ordering to the demands of the immediate scene, which may favour an order that departs from the stable size-first preference.

Referential usefulness also acts before an ordering choice becomes observable. Models of reference planning propose that the contrast set shapes which properties are selected for mention (Fukumura, Van Gompel, and Pickering, 2010; Van Gompel et al., 2019). The inclusion of redundant material reflects both the processes through which a description is planned and its usefulness to a listener. Speaker-internal accounts of reference production, often termed *egocentric*, propose that readily accessible properties can be selected before the speaker has completed the search for a sufficient description (Pechmann, 1989; Engelhardt, Bailey, and Ferreira, 2006). Audience-oriented accounts assign the additional material a cooperative role: redundant properties can guide visual search and hedge uncertainty about which cue the listener will use (Rubio-Fernández, 2016; Rubio-Fernandez, 2019; Degen et al., 2020; Tourtouri, Delogu, and Crocker, 2021). In a shared visual scene, these routes can converge because a property that is easy for the speaker to retrieve may also help the listener identify the target. The contribution of these pressures to ordering depends on which properties the speaker includes in the description.

Slider ratings and direct production reveal different aspects of this dependence. A one-adjective utterance can resolve reference without creating an order contrast. Slider ratings hold adjective selection and utterance length fixed by presenting two complete orders. They establish an explicit ordering preference and leave open how often production reaches that choice. Direct production reveals how referential context affects the selection of adjectives as well as their order when selected together. Comparing the tasks therefore relates an explicit preference between orders to the circumstances in which speakers produce those adjectives together.

2.3 Incremental RSA and the Full Production Distribution

The Rational Speech Act (RSA) framework models speaker choice in terms of an utterance's usefulness to a listener and its production cost (Frank and Goodman, 2012; Goodman and Frank, 2016; Degen, 2023). It allows assumptions about semantic interpretation and speaker choice to be specified separately within a common account of reference. An ordering preference can consequently depend both on how the adjectives are interpreted and on which alternatives compete during production. Standard RSA models treat complete utterances as the units of speaker choice. The same pragmatic reasoning can be defined incrementally, with the speaker choosing one word at a time relative to the alternatives available after the words already selected (Cohn-Gordon, Goodman, and Potts, 2019). The alternatives available at each step can affect adjective selection and the decision to continue or stop, with consequences for the distribution of complete utterances. Pragmatic models of reference capture related asymmetries in modifier position, order, and redundancy (Degen et al., 2020; Rubio-Fernández, Mollica, and Jara-Ettinger, 2021). Variation in planning scope further motivates considering how complete-utterance evaluation and incremental choices may jointly influence production (Griffin and Bock, 2000; Ferreira and Swets, 2002; Konopka and Meyer, 2014). An utterance-level plan can constrain the intended description while alternatives available at each step continue to influence the next word. In the present *plan-guided incremental* account, this relation is represented by allowing both complete-utterance utility and competition during incremental choice to contribute to the probability of the completed utterance.

Schlotterbeck and Wang (2023) introduced the slider paradigm on which Studies 1a and 1b build. Their qualitative RSA model combined sequential context updating within hierarchical semantic composition with incremental speaker choice. Together, these mechanisms captured a stable size-first preference and its modulation by discriminatory strength. The analysis considered two fixed size–colour orders and left the separate predictive contributions of the two directions unresolved. The present study extends that starting point with a direct slider replication and a production distribution over one-, two-, and three-adjective utterances, allowing adjective selection, utterance length, and order to be modelled jointly. This extension makes it possible to examine whether semantic updating and incremental adjective choice improve prediction of the descriptions speakers actually select.

3 Behavioural Experiments

The three studies expose different aspects of the same empirical problem. Study 1a and its direct replication, Study 1b, ask participants to compare two completed size–colour orders using sliders. Study 2 elicits descriptions through direct production, allowing adjective selection and utterance length to vary along with order. The two tasks therefore connect explicit ordering preferences to the choices that make an order observable in production. Referential context varies whether size, colour, or both are needed to identify the target, manipulating the relative usefulness of the two properties (Fukumura, 2018; Rubio-Fernandez and Jara-Ettinger, 2020; Rubio-Fernández, Mollica, and Jara-Ettinger, 2021). Size discriminability separately varies how clearly size differences can be perceived. Crossing these factors allows us to ask whether the influence of perceptual reliability depends on which properties support reference, and how these pressures affect ordering judgements and production. Because size is gradable, this comparison also bears on the role of the comparison class in size interpretation (Scontras, Degen, and Goodman, 2019; Schlotterbeck and Wang, 2023).

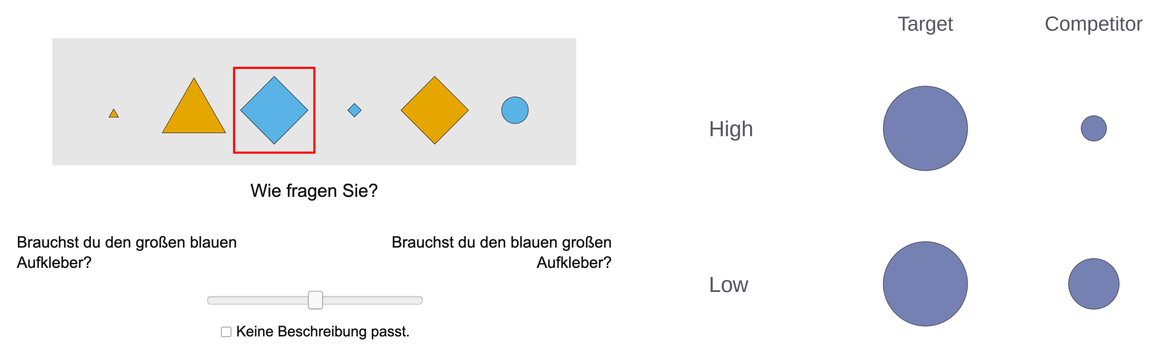
The directional hypotheses connect these theoretical motivations to specific behavioural contrasts. For slider judgements, greater referential usefulness of size predicts stronger size-first preference, yielding the ordering size-sufficient > both-necessary > colour-sufficient; a stable size-first preference predicts ratings above neutrality even when colour alone identifies the target. For production, the targeted interaction contrasts evaluate whether lower size discriminability increases redundancy more strongly when size alone is sufficient, and whether higher discriminability increases size-initiality more strongly when both properties are necessary. The slider interaction contrasts assess how discriminability effects differ across contexts, with Study 1b testing whether the pattern observed in Study 1a recurs.

3.1 General Method Shared between Experiments

3.1.1 Design

The three studies shared the same basic factorial design and visual displays, with different response formats. On every critical trial, participants saw a six-object display with one framed target. The target was always the largest object that matched the intended colour and form. Referential context varied within participants: size alone, colour alone, or both properties were needed to identify the target. Size discriminability varied between participants, so that each participant experienced either more pronounced or less pronounced target–competitor size differences throughout the task (Figure 2). The behavioural comparisons below focus on size–colour trials. The production study also included size–form and colour–form trials, which enter the later computational model comparison.

A B



C

Design dimension	Levels
Referential context	Size sufficient; both necessary; colour sufficient (within participants)
Size discriminability	High; low (between participants)

Figure 2: Experimental design and materials. Panel A reproduces an exact slider-trial screenshot: the framed object is the target, and participants rated the two German adjective orders. Panel B illustrates the size-discriminability manipulation schematically using recorded stimulus levels: the target–competitor size gap is larger under high discriminability and smaller under low discriminability. Panel C summarises the factorial design for the size–colour comparisons.

3.1.2 Materials and procedure

All studies were conducted online via Prolific. Compensation was set slightly above the platform's median level. The task was framed as a reference game about collectible stickers: participants gave or evaluated a description for a partner who saw the same stickers in a different order and lacked the target frame. This made spatial terms uninformative and kept the task focused on adjective descriptions. The noun was always *Aufkleber* ('sticker'), and the materials varied the objects' size, colour, and form across counterbalanced lists. Practice trials familiarised participants with the task, and main-trial order was randomised within blocks. Stimulus construction, trial counts, exclusions, and annotation guidelines appear in Appendix A.

3.2 Study 1a: Slider-Rating Study

3.2.1 Participants

Study 1a uses the same dataset reported by Schlotterbeck and Wang (2023). The retained sample comprised 134 participants (70 in the low- and 64 in the high-discriminability group), recruited via Prolific and reporting German as their native language. Exclusions addressed missing records, extreme completion times, and failures on control or task-relevance checks. The size–colour analysis contained 3,485 observations.

3.2.2 Procedure

On each critical trial, two competing adjective orders appeared on opposite sides of a continuous slider, together with a "none of the descriptions fits" option. Left–right presentation of the two orders was randomised across trials. Ratings were centred so that zero indicated neutrality, positive values favoured size-first, and negative values favoured colour-first order, on a scale from -0.5 to 0.5.

3.2.3 Results

The Bayesian hierarchical analysis revealed a credible interaction between size discriminability and referential context in the planned comparison of size-sufficient and both-necessary displays (Figure 3). The high-minus-low discriminability effect was smaller in the both-necessary context than in the size-sufficient context (posterior median difference $= -0.085$, 95% credible interval [CrI] $[-0.151, -0.022]$; posterior probability .995 of a negative contrast). Higher discriminability was associated with a shift towards size-first ratings in size-sufficient displays and towards colour-first ratings in both-necessary displays, although the credible intervals for both within-context effects included zero. The corresponding interaction contrast between colour-sufficient and size-sufficient displays was less conclusive (difference $= -0.063$, 95% CrI $[-0.172, 0.039]$; posterior probability .885 of a negative contrast). Thus, the evidence for an interaction concerned how the discriminability effect differed between contexts; the direction of each individual effect was less precisely estimated. Model estimates in the experimental results are reported as posterior medians, and slider contrasts are expressed in centred-rating units. Directional posterior probabilities quantify the probability of the stated contrast direction under the fitted model. Model specifications appear in Appendix A.

Averaged across discriminability conditions, the analysis also revealed a credible effect of referential context. Size-first ratings were higher in size-sufficient than in both-necessary displays (difference $= 0.094$, 95% CrI $[0.061, 0.127]$), and higher in both-necessary than in colour-sufficient displays (difference $= 0.171$, 95% CrI $[0.131, 0.208]$). The joint posterior probability of the predicted context ordering exceeded .999, supporting the hypothesis that size-first preference increases with the referential usefulness of size. Even when colour alone identified the target, the estimated rating remained slightly above neutrality (0.054, 95% CrI $[0.003, 0.109]$; posterior probability .982 of a positive rating). Referential usefulness thus modulated the strength of the size-first preference, which persisted across all three contexts.

3.3 Study 1b: Direct Slider Replication

3.3.1 Participants and procedure

The replication used the same task and materials with an independent participant sample. Applying the Study 1a exclusion criteria retained 104 of 132 participants and 2,786 size–colour observations.

3.3.2 Results

The same Bayesian hierarchical analysis did not yield clear evidence for an interaction between size discriminability and referential context in Study 1b (Figure 3). Both planned interaction contrasts had credible intervals spanning zero: the both-necessary minus size-sufficient contrast (difference $= -0.014$, 95% CrI $[-0.097, 0.071]$) and the colour-sufficient minus size-sufficient contrast (difference $= 0.034$, 95% CrI $[-0.082, 0.157]$). The posterior probabilities of negative contrasts, corresponding to the directions estimated in Study 1a, were .623 and .285, respectively. The interaction observed in Study 1a was therefore not clearly replicated.

The effect of referential context was replicated. Averaged across discriminability conditions, size-first ratings were higher in size-sufficient than in both-necessary displays (difference $= 0.073$, 95% CrI $[0.029, 0.115]$), and higher in both-necessary than in colour-sufficient displays (difference $= 0.140$, 95% CrI $[0.090, 0.190]$). The joint posterior probability of the predicted context ordering was .999. The preference in colour-sufficient displays again remained slightly above neutrality (0.057, 95% CrI $[0.001, 0.116]$; posterior probability .977 of a positive rating).

Both studies therefore showed the same graded relation between referential usefulness and ordering preference, including a residual size-first preference when colour alone identified the target.

The estimated effect of higher size discriminability was positive in all three contexts. Within-context comparisons supported higher size-first ratings in size-sufficient displays (difference $= 0.090$, 95% CrI $[0.014, 0.160]$) and colour-sufficient displays (difference $= 0.124$, 95% CrI $[0.016, 0.240]$); the interval for both-necessary displays included zero. This pattern provides evidence that clearer size differences can strengthen size-first preference, while leaving uncertainty about variation in that effect across referential contexts.

Figure 3: Ordering judgements in Study 1a and Study 1b by referential context and size discriminability. Points show posterior medians of expected centred ratings for a new participant; error bars show 95% credible intervals. Positive values indicate size-first preference; zero indicates neutrality. The two studies use the same model specification and interval definition.

3.4 Study 2: Production Study

3.4.1 Participants

The production study retained 113 participants (60 in the low- and 53 in the high-discriminability group), recruited via Prolific; participants from either slider study were ineligible. Exclusions addressed extreme completion times, missing or unannotatable responses, and more than 80% identical responses. The latter criterion selects a sample that varies its descriptions across trials. The behavioural comparisons use 3,034 size–colour trials; the later computational comparison uses all 9,100 annotated trials across the three adjective families.

3.4.2 Procedure

Participants saw the framed target together with the fixed sentence frame *den ___ Aufkleber* and typed the adjective material before the noun. Practice trials introduced the relevant colour and form vocabulary. The first author annotated the responses following guidelines that preserve adjective content and order. Each adjective was assigned to the size, colour, or form class, and the sequence of these classes defined one of 15 utterance categories: three for one-adjective, six for two-adjective, and six for three-adjective utterances. Misspellings and transparent lexical alternatives were normalised when their semantic class was unambiguous. Spatial descriptions, comparative or ordinal size utterances, and responses that could not be assigned to a target adjective class were excluded from the production model comparison. The complete annotation scheme appears in Appendix A.

The annotated responses were further classified for the two experimental hypotheses. *Redundant adjective use* indicates a response containing more properties than needed for unique reference: multiple adjectives in a one-property-sufficient context, or three adjectives when both properties are needed. *Size-initiality* indicates that a response begins with size, including a response consisting of size alone. It therefore reflects adjective selection as well as order. Order conditional on producing both size and colour is described separately. The two outcomes were analysed with Bayesian hierarchical models containing referential context, size discriminability, and their interaction; the full specifications and planned contrasts appear in Appendix A.

3.4.3 Results

Redundant adjective use. The Bayesian hierarchical analysis revealed a credible interaction between size discriminability and referential context in redundant adjective use (Figure 5, Panel A). The increase in redundancy under low discriminability was larger in size-sufficient displays than its average in the other two contexts (difference $= 15.3$ percentage points, 95% CrI $[7.2, 23.6]$; posterior probability $> .999$ in the hypothesised direction). Within size-sufficient displays, lower discriminability increased the estimated probability of redundant adjective use by 23.4 percentage points (95% CrI $[15.1, 31.8]$). The corresponding increases in both-necessary and colour-sufficient displays were smaller, and their credible intervals included zero. These results supported the hypothesis that reduced size discriminability particularly increases redundant adjective use when size alone can identify the target.

The observed response distribution further indicated that referential context shaped adjective content and utterance length (Figure 4). Both-necessary scenes generally elicited longer descriptions, whereas one-property-sufficient scenes more often elicited a single adjective. Colour-sufficient scenes were commonly resolved with colour alone, so no size–colour ordering decision was observed on those trials. Referential context therefore affected how often speakers produced the two adjectives whose order was compared in the slider task.

Initial adjective and conditional order. The analysis of size-initiality also revealed a credible interaction between size discriminability and referential context (Figure 5, Panel B). The high-minus-low discriminability effect was larger in both-necessary displays than its average in the two one-property-sufficient contexts (difference $= 16.9$ percentage points, 95% CrI $[7.7, 25.6]$; posterior probability $> .999$ in the hypothesised direction). Within both-necessary displays, higher discriminability increased the estimated probability of a size-initial response by 14.1 percentage points (95% CrI $[3.6, 24.2]$). The effect in size-sufficient displays was close to zero and uncertain, whereas colour-sufficient displays showed a small decrease of 3.2 percentage points (95% CrI $[0.6, 6.8]$). These results supported the hypothesis that higher size discriminability particularly favours size-initial responses when both properties are needed to identify the target.

Averaged across discriminability conditions, size-initial responses were more probable in size-sufficient than in both-necessary displays (difference $= 19.3$ percentage points, 95% CrI $[13.9, 24.8]$), and more probable in both-necessary than in colour-sufficient displays (difference $= 53.1$ percentage points, 95% CrI $[45.9, 60.1]$). Speakers were therefore more likely to begin with size when it was more useful for reference, including when they used size as the only adjective.

Conditional on producing both size and colour, size-first descriptions remained more frequent, and colour-first descriptions were comparatively rare. This conditional preference describes responses containing both properties, while size-initiality also includes the decision to use size alone. The production results therefore locate context sensitivity in adjective selection, length, and initial position, alongside a descriptive size-first preference among responses containing both adjectives.

3.5 Interim Discussion

The studies provided different patterns of evidence for the interaction between size discriminability and referential context. Study 1a showed the clearest interaction between size-sufficient and both-necessary displays, while Study 1b did not provide clear evidence for the same pattern. In production, lower size discriminability increased redundant adjective use most clearly when size alone was sufficient; higher discriminability increased size-initial responses most clearly when both properties were necessary. Perceptual reliability therefore affected different aspects of a description depending on the referential demands of the scene.

The two slider studies consistently showed a referential-context gradient: size-first preferences were strongest when size identified the target and weakest when colour did, with a small positive preference remaining in colour-sufficient displays. Direct production revealed how the same contextual demands also shaped adjective selection and utterance length. When one property identified the target, speakers frequently produced a single adjective, leaving no order among multiple adjectives to observe. The context manipulation can therefore change explicit judgements between completed orders and the likelihood that production includes those adjectives together.

Production also revealed substantial use of three-adjective utterances and pronounced participant variation in how often these longer descriptions occurred. These responses extend beyond the material required for unique reference and involve several adjective choices. Modelling these data therefore requires assigning probability to descriptions of different lengths and contents, including the redundant descriptions whose frequency varies across

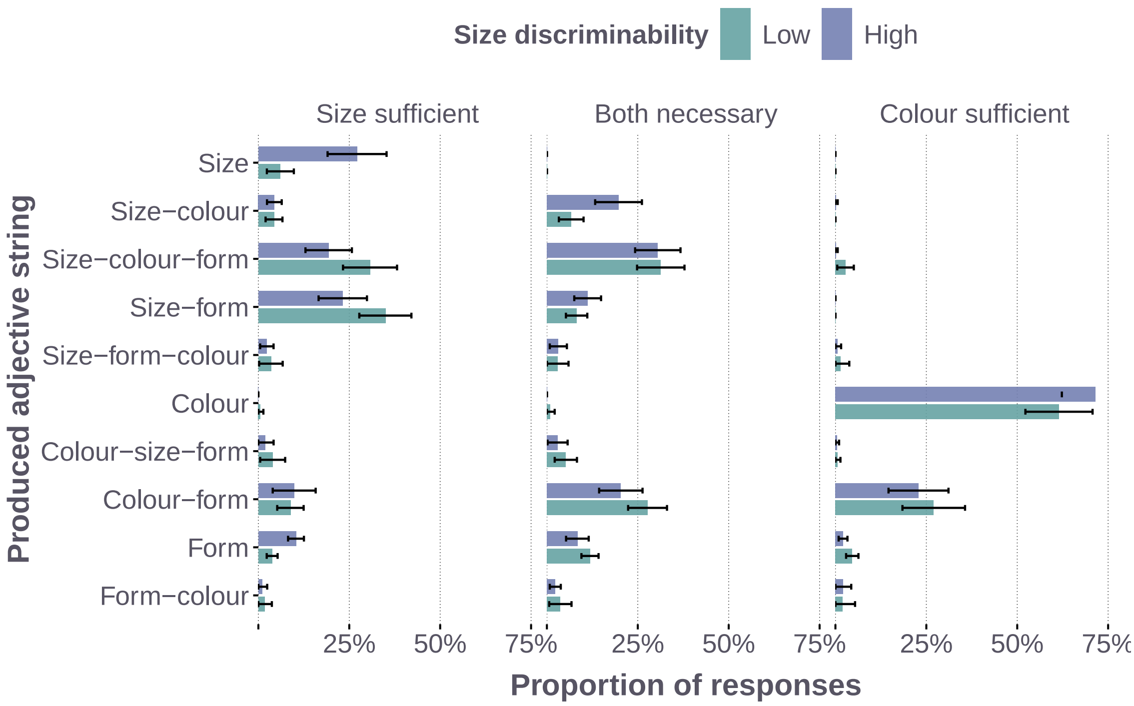


Figure 4: Ordered response distribution across the 3,034 size–colour production trials. Bars show mean participant-level proportions by referential context and size discriminability, with 95% confidence intervals. Adjective labels preserve surface order; the figure shows responses reaching at least 2% in one condition.

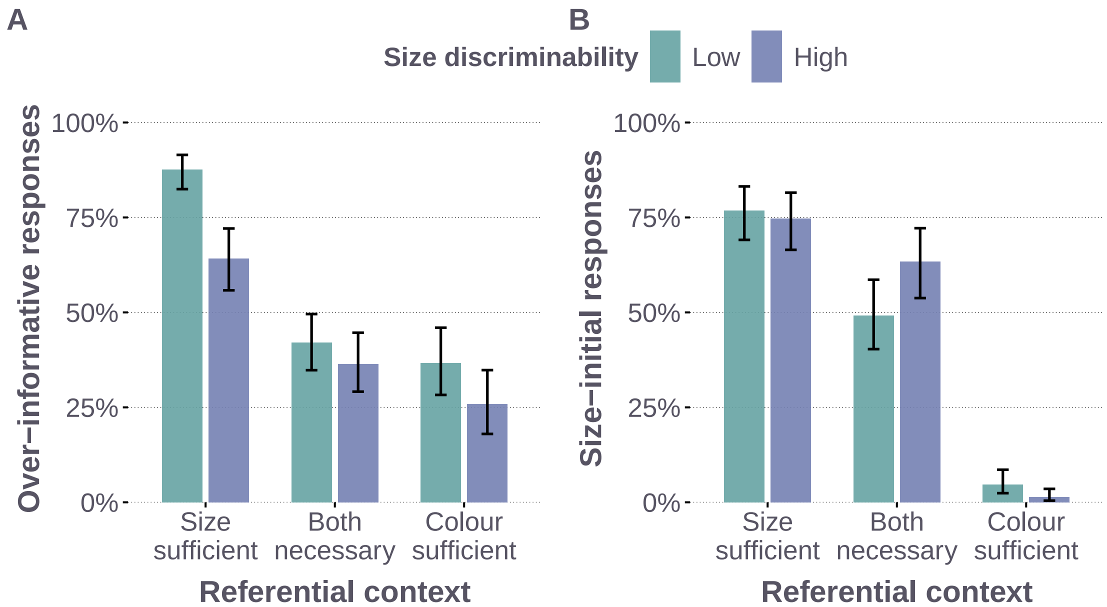


Figure 5: Redundant adjective use and size-initial responses in size–colour production. Panel A shows redundant adjective use: a multi-adjective response in a one-property-sufficient context or a three-adjective response when both properties are necessary. Panel B shows the probability that a response begins with size. Bars show Bayesian posterior medians and error bars show 95% credible intervals. Size-initiality combines adjective selection and order.

participants. The following framework represents these alternatives within a common utterance space, allowing us to compare how semantic interpretation and incremental choice contribute to their use.

4 Formal Modelling Framework

The formal framework of the current paper builds on the central intuition of the Rational Speech Act (RSA) model: rational communication links comprehension and production, such that a listener interprets an utterance to infer the speaker's intended referent and a pragmatic speaker assigns greater probability to utterances that guide the listener successfully to the target (Frank and Goodman, 2012). In the vanilla RSA model, the speaker evaluates a fixed set of completed utterances according to their communicative efficiency. The present model extends the vanilla RSA model in three respects. First, it expands the response space to all fifteen ordered one-, two-, and three-adjective utterances available in the experiment, bringing adjective selection, utterance length, and ordering into a single production distribution. This extends the two fixed size–colour orders modelled by Schlotterbeck and Wang (2023) to the richer structure of adjective use observed in direct production. Second, it allows recursive updating of the listener state as adjectival meanings compose from the noun outwards. Third, it extends the speaker to incremental production by allowing adjective choices to be evaluated incrementally along the surface linear sequence.

These extensions allow the contributions of semantic interpretation and incremental adjective choice to be evaluated within the complete production distribution. The production architecture determines how strongly competition during incremental adjective choice contributes to the probability of the completed utterance, ranging from global evaluation through plan-guided production to fully incremental adjective choice. The semantic regime determines whether interpreting an adjective can change the comparison class for subsequent hierarchical composition.

4.1 Utterance Space, State Space, and the Literal Listener

The model assigns probabilities to all fifteen ordered strings formed from size, colour, and form: three one-adjective utterances, six two-adjective utterances, and six three-adjective utterances. This response space allows the model to predict which properties speakers mention, how many adjectives they produce, and how jointly selected adjectives are ordered. All models use the observed six-object display. Size has a graded meaning defined relative to a comparison class (Schmidt et al., 2009; Franke, Scontras, and Simonic, 2019). Colour provides a graded target-matching cue. Form gives equal literal support to matching and nonmatching objects; its use is supported through salience and task-specific production terms. For every complete or partial adjective string, the literal listener L applies adjectival modification from the noun outwards, corresponding to right-to-left composition in the prenominal strings considered here.

Let $u_{\leq t} = w_1 \dots w_t$ be a surface prefix comprising the first t adjectives, and $P_0(o \mid X) = 1/6$ the uniform prior over the six objects in display X. Under context-fixed semantics, the literal listener after this prefix is

$$L(o \mid u_{\leq t}, X) \propto P_0(o \mid X) \prod_{j=1}^{t} [\![w_j]\!]_{\text{fix}}(o). \tag{1}$$

The listener combines the meanings made available by the surface prefix and assigns probability to every object that remains compatible with them. The probability assigned to the target provides the communicative value used by all three production accounts.

4.2 Incremental Adjective Choice and Plan-Guided Production

The three production accounts share an *incremental evaluation* rule: the speaker evaluates how the adjectives in an utterance guide the listener to the target at each adjective position. Accumulating these evaluations allows identification during the description to contribute to its utility. The choice architecture determines how this utility enters the competition among alternatives: global choice compares complete utterances, while incremental choice includes local competition among adjectives at each position. At position t, the utility of candidate adjective a for target o^* is

$$U_t(a \mid u_{<t}, o^*, X) = \alpha \log L(o^* \mid u_{<t} + a, X) + \lambda_{\text{salience}} z_a, \tag{2}$$

where α controls sensitivity to listener success and z_a represents the visual salience of the candidate property. For a completed utterance u, the model combines the accumulated utility of its selected adjectives with a term for competition among alternatives at each adjective position:

$$U^{\text{complete}}(u, o^*, X) = \sum_t U_t(w_t \mid u_{<t}, o^*, X), \tag{3}$$

$$\mathcal{N}^{\text{local}}(u, o^*, X) = \sum_t \log \sum_{a \in \mathcal{A}(u_{<t})} \exp U_t(a \mid u_{<t}, o^*, X), \tag{4}$$

where $\mathcal{A}(u_{<t})$ contains the adjective types still available after the current prefix. The complete-utterance utility U^{complete} is the sum of these incremental evaluations and is shared by all three production architectures. The local-normalisation term $\mathcal{N}^{\text{local}}$ accumulates the log denominators of the local softmax choices over adjectives available at each position. In a locally normalised choice, a selected adjective receives lower probability when several alternatives at the same prefix also have high utility. Two completed utterances with similar U^{complete} can therefore receive different probabilities because they pass through prefixes with different degrees of competition.

The probability of a completed utterance is

$$S_\kappa(u \mid o^*, X) \propto \exp\left\{U^{\text{complete}}(u, o^*, X) - \kappa\mathcal{N}^{\text{local}}(u, o^*, X) + \beta\Omega(u) + T(u, X)\right\}, \tag{5}$$

The parameter κ determines how strongly competition during incremental adjective choice contributes to the probability of a completed utterance. With $\kappa = 0$, the local normalisers drop out and the *global* speaker normalises over the fifteen completed utterances. With $\kappa = 1$, the *fully incremental* speaker retains the complete contribution of locally normalised adjective choices. An interior value defines the *plan-guided* speaker: complete-utterance utility and competition among locally available adjectives jointly determine the response probability. The global model therefore combines incremental evaluation with global choice among complete utterances.

The remaining terms capture stable order expectations through $\beta\Omega(u)$ and task-specific pressures on adjective selection and utterance length through $T(u, X)$, which we refer to as the *task link*. They extend the informativity–cost trade-off of a vanilla RSA speaker: target-directed utility is balanced against conventional order preferences and task-specific production costs. These terms are shared across the three production accounts. Length choice enters through the comparison of completed responses and $T(u, X)$.

Before the final Laplace smoothing, the same model can be written at common parameter values as a geometric interpolation between the global and fully incremental endpoints:

$$S_\kappa(u \mid o^*, X) \propto S_{\text{G}}(u \mid o^*, X)^{1-\kappa} S_{\text{I}}(u \mid o^*, X)^{\kappa}, \tag{6}$$

where S_{G} and S_{I} are the endpoint distributions evaluated with the same listener, stable-order component, task link, and parameter values. The proportionality in Equation (6) renormalises the interpolated scores over the fifteen completed utterances.

The full parameterisation and response likelihood appear in Appendix B; Appendix F derives the local-normalisation term and its relation to the model endpoints.

4.3 Sequential Context Updating within Hierarchical Semantic Composition

The semantic comparison concerns the inner-to-outer direction of hierarchical composition. Let $c_1, \dots, c_T$ denote the modifiers in a completed utterance ordered from the noun outwards, so that $c_1 = w_T$ for prenominal modification. Starting from $C_0 = P_0$, composition restricts the context state available to the literal listener at each step:

$$C_j(o) \propto C_{j-1}(o)\, [\![c_j]\!]_{\rho_j}(o), \qquad j = 1, \dots, T, \tag{7}$$

where $[\![c_j]\!]_{\rho_j}$ is the modifier's meaning under the selected semantic regime.

The semantic regime determines the comparison class used for size. Under context-fixed semantics, the graded size threshold remains anchored to the original scene. Under sequential context updating, the threshold is recomputed from the context state left by the hierarchically prior modifier. When this restriction makes the target more distinctive in size, sequential context updating can give the size-first order a referential advantage. The literal listener applies this noun-outward computation to each available surface prefix, linking semantic interpretation to the adjective utilities defined above for each position. The comparison therefore asks whether recomputing the size comparison class adds predictive accuracy in the observed production data. The simulation examines how this operation contributes to an ordering advantage across production models and a broader range of contexts. The threshold construction and further semantic details appear in Appendices B and G.

5 Model Comparison and Predictive Evidence

The model comparison tests how hierarchical semantic composition and incremental adjective choice contribute to the observed distribution of utterances. Global, plan-guided, and fully incremental production are each paired with context-fixed semantics or sequential context updating in a matched 3×2 comparison. We then assess whether participant variation in incremental adjective choice improves prediction and examine which production patterns the resulting model captures.

5.1 Inference and Model Comparison Setup

We fit each model to the same 9,100 production trials from 113 participants using Bayesian inference in NumPyro (Phan, Pradhan, and Jankowiak, 2019). All six models share the same fifteen-utterance response space, task link, semantic parameters, and participant hierarchies for pragmatic optimality α_i and stable-order weight β_i. They differ in production architecture, specified by fixing $\kappa = 0$, estimating one shared κ, or fixing $\kappa = 1$, and in whether the interpretation of size is recomputed during hierarchical composition. The comparison estimates the predictive contribution of each component while holding the remaining specification constant. The subsequent participant refinement estimates individual incremental-choice weights κ_i within the plan-guided architecture.

Figure 6 separates fixed theoretical choices, estimated parameters, participant variation, and observed responses. The fixed variables A and R index production architecture and semantic regime. For participant i on trial t, the model maps the display and target X_{it} and the stable-order score $\Omega(u)$ to a probability $\pi_{it}(u)$ over the fifteen possible utterances, from which the observed response y_{it} is drawn. The shared coefficients govern visual salience, adjective selection, and utterance length.

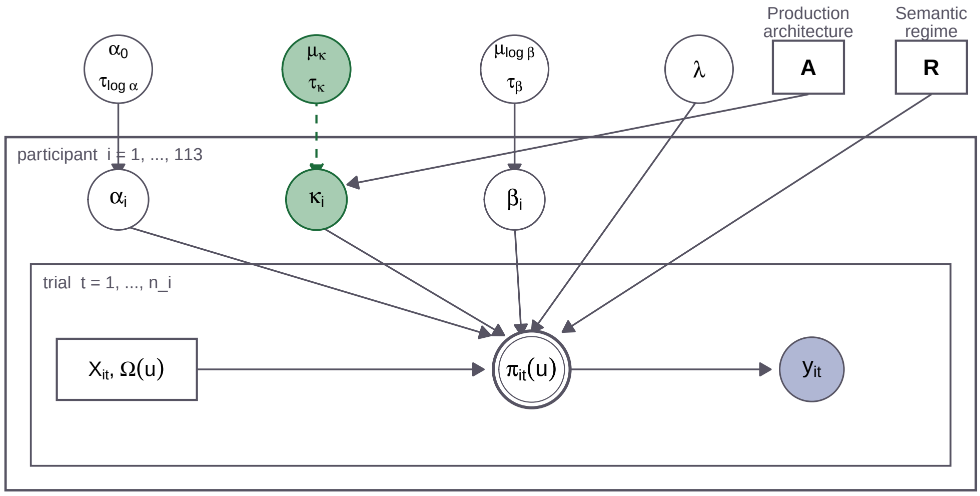


Figure 6: Hierarchical structure of the production model and the restrictions used in the matched comparisons. Solid circles represent estimated quantities, squares represent fixed model or trial inputs, the double circle represents the deterministic utterance distribution, and the shaded circle represents the observed response. The dashed κ hierarchy applies to the plan-guided model with participant-specific incremental-choice weights.

Approximate leave-one-out expected log predictive density (ELPD) measures how much probability each model assigns to held-out responses (Vehtari, Gelman, and Gabry, 2017). Higher ELPD indicates better prediction of the complete utterance distribution. The predictive target is another response from a participant represented in the dataset, with that participant's remaining trials available to estimate their participant-specific parameters. Model development used these data, and response-wise leave-one-out prediction evaluates the specifications selected through that development. Uncertainty in paired model comparisons is estimated with a participant-level Bayesian bootstrap and reported as 95% credible intervals. The same predictive target and uncertainty calculation are used throughout the architecture and hierarchy comparisons. The model specification, inference procedure, and numerical diagnostics appear in Appendices B and C.

5.2 Model-Comparison Results

Production architecture. For the second research question, the Bayesian model comparison favoured plan-guided production over both global and fully incremental production (Figure 7, Panel A). Averaged across semantic regimes, the plan-guided model had higher predictive accuracy than the global model (ΔELPD $= 256.9$, 95% CrI $[71.8, 404.7]$) and the fully incremental model (ΔELPD $= 548.5$, 95% CrI $[349.8, 795.1]$). These contrasts supported the plan-guided production hypothesis that complete-utterance evaluation and competition during incremental adjective choice jointly shape the response distribution. In the context-fixed model, the posterior mean shared incremental-choice weight was $\kappa = .446$, with a 95% CrI of $[.407, .486]$ (Panel B). The estimate lay between the global and fully incremental endpoints, indicating an intermediate contribution of local competition within the fitted architecture.

The predictive differences were most apparent in form-involving trials, where colour and form competed for the initial position. Colour-initial responses accounted for 81.7% of the plan-guided model's ELPD advantage over the average of the global and fully incremental models. Different opening adjectives leave different continuations available, allowing competition at later positions to affect the probabilities of the completed descriptions. The advantage therefore extended beyond the motivating size–colour contrast to the selection and ordering of properties in the broader response space.

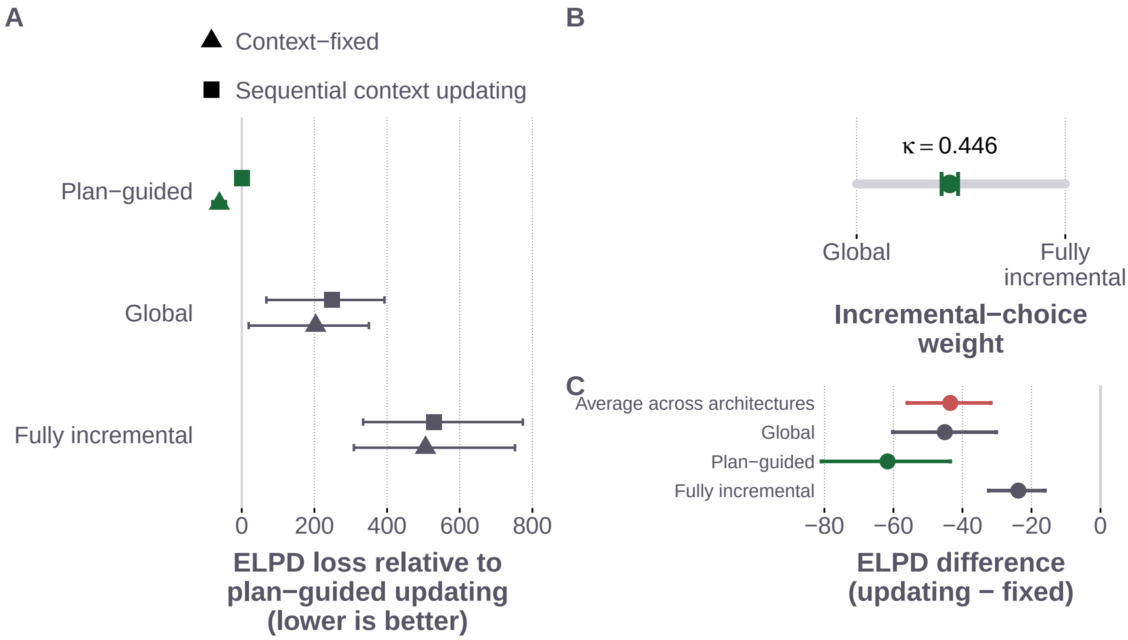


Figure 7: Matched comparisons of production architecture and sequential context updating. Panel A reports predictive loss for all six matched models relative to the plan-guided model with sequential context updating; lower values indicate better leave-one-out prediction. Panel B places the shared incremental-choice weight κ between the global and fully incremental endpoints. Panel C shows updating-minus-fixed ELPD for each production architecture and their average. Intervals in Panels A and C are 95% credible intervals from the participant-level Bayesian bootstrap; Panel B shows the 95% posterior credible interval for κ. The vertical line in Panel C marks zero.

Hierarchical semantic composition. The third research question asked whether updating the size comparison class improves prediction of production. The comparison favoured context-fixed semantics over sequential context updating under all three production architectures (Figure 7, Panel C). The average updating-minus-fixed contrast was negative (ΔELPD $= -43.5$, 95% CrI $[-56.0, -31.8]$), indicating a credible predictive advantage for the context-fixed models. This advantage remained when incremental-choice weights were allowed to vary across participants, as described below.

Decomposing the participant-specific model comparison showed that the semantic difference chiefly concerned how much speakers said and which properties they mentioned. Context-fixed semantics improved predictions of utterance length (ΔELPD $= 21.5$, 95% CrI $[12.4, 30.9]$) and adjective content conditional on length

(ΔELPD $= 31.1$, 95% CrI $[23.5, 39.2]$). For order conditional on adjective content, the estimated advantage was small and its interval included zero (ΔELPD $= 1.5$, 95% CrI $[-1.8, 4.9]$). The comparison therefore did not establish a clear predictive advantage for either semantic regime in conditional order.

The response residuals illustrated these differences in size-sufficient size–form trials. When size discriminability was high, both models overpredicted three-adjective size–colour–form responses: the observed proportion was 21.1%, compared with 27.3% under fixed semantics and 31.2% under updating. This response showed the largest change in absolute residual error between the models. In the same condition, both models underpredicted size–form utterances, with a larger discrepancy under updating. Updating improved predictions for some responses, including size-only descriptions, but its overall disadvantage was accompanied by larger errors in the relative frequencies of these two- and three-adjective utterances. Full contrasts, the score decomposition, and the residual analysis appear in Appendix C.3.

5.3 Plan-Guided Production and Participant Variation

Allowing incremental-choice weights to vary across participants yielded a credible improvement over the otherwise matched model with a shared weight. Under context-fixed semantics, the participant-specific model gained 364.7 ELPD (95% CrI $[248.2, 521.0]$); the gain under sequential context updating was similar. The posterior mean population location remained intermediate between the global and fully incremental endpoints ($\mu_\kappa = .472$, 95% CrI $[.337, .618]$), alongside substantial variation in participant-specific weights. Thus, the improvement came from accommodating differences across participants while retaining an intermediate population location.

The predictive benefit was concentrated in two- and three-adjective utterances, whereas predictions of single-adjective responses became less accurate. The gains for longer utterances outweighed this loss and extended to three-adjective descriptions, which contain redundant information in the present task. Participant variation therefore improved the model's account of descriptions involving several adjective choices, including the selection and ordering of additional properties. Appendix E reports the response-length contrasts and participant-level diagnostics.

5.4 Predictive Adequacy of the Best-Predicting Model

Posterior predictive checks returned to the first research question by assessing how closely the model reproduced the observed context sensitivity of adjective selection, utterance length, and initial position. The plan-guided model with context-fixed semantics and participant-specific incremental-choice weights captured the broad context-dependent shifts in size-initial responses and redundant adjective use (Figure 8, Panel A). The comparison across all fifteen utterances and eighteen experimental conditions showed correspondence between observed and predicted proportions, alongside discrepancies for particular responses (Panel B).

The condition-wise checks made these patterns more explicit (Figure 9). In size–colour trials, the model reproduced the predominance of colour-only responses when colour was sufficient and the greater use of size alone when size was sufficient and highly discriminable. It nevertheless overpredicted colour-only responses in colour-sufficient conditions. In size-sufficient and both-necessary conditions, it tended to overpredict size–form responses and underpredict size–colour–form responses. These discrepancies indicate that the model captured the broad changes in descriptive content while misestimating the balance between some two- and three-adjective utterances.

A further discrepancy occurred in size–form trials when both properties were necessary and size discriminability was high. The model underpredicted size-initial responses (predicted: 45.2%; observed: 59.5%) and overpredicted form-initial responses (predicted: 13.5%; observed: 4.2%). The checks therefore identified remaining errors in initial-adjective choice as well as in the distribution of completed utterances. Descriptive measures of fit appear in Appendix C.

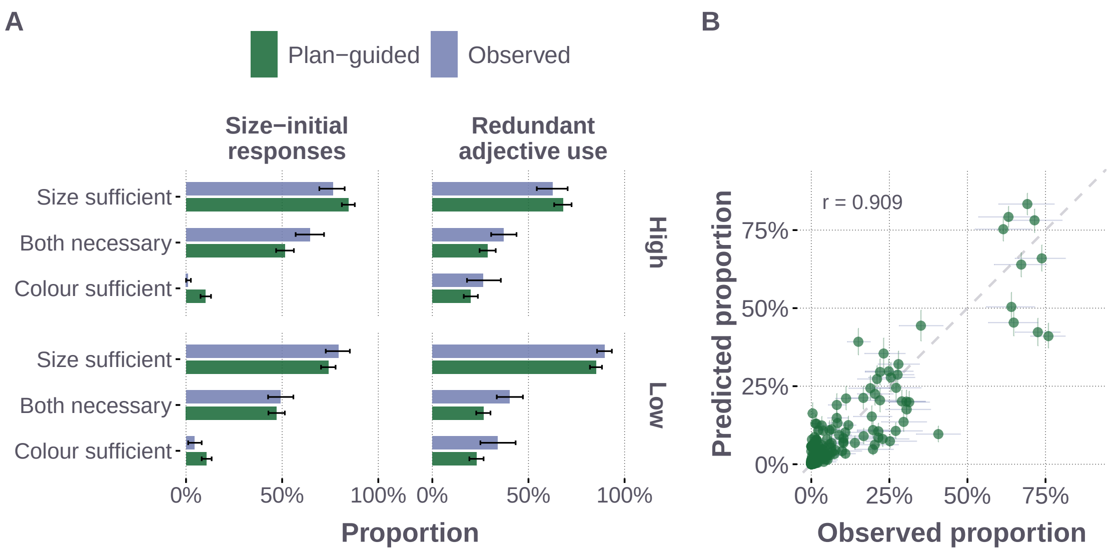


Figure 8: Posterior-predictive adequacy of the best-predicting plan-guided model. Panel A compares observed and posterior-predicted size-initial responses and redundant adjective use in size–colour trials, with referential context on the vertical axis and size discriminability in facet rows. Intervals show participant-bootstrap 95% intervals for the observed proportions and 95% posterior-predictive intervals for the model predictions. Panel B compares all 270 response cells; horizontal intervals show participant-bootstrap uncertainty in the observations, vertical intervals show 95% posterior-predictive uncertainty, and the dashed line marks equality.

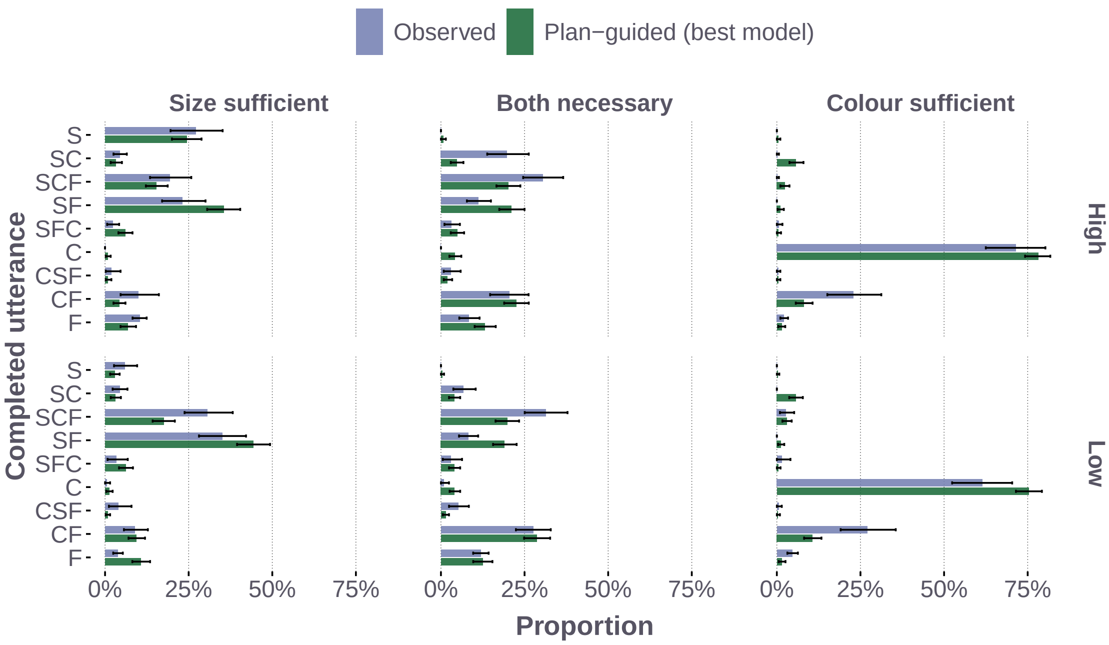


Figure 9: Observed and posterior-predicted proportions for frequent utterances in size–colour trials under the best-predicting plan-guided model. The figure includes utterances whose observed or mean posterior-predicted proportion reaches 5% in at least one context-by-discriminability condition, with proportions pooled across trials within each condition. Intervals show participant-bootstrap 95% intervals for observed proportions and 95% posterior-predictive intervals for model predictions. Adjective labels preserve surface order: S denotes size, C colour, and F form.

These predictive comparisons leave open how a communicative advantage for size-first order arises under each semantic regime. The following simulation examines how this advantage arises and how it changes when adjective selection and utterance length enter the speaker's alternatives.

6 Simulation: Conditions for a Size-First Advantage

A communicative account of adjective ordering seeks to explain why one order can be more useful for identifying a referent than another. Earlier Monte Carlo simulations found an average referential advantage for orders that place more subjective adjectives farther from the noun, motivating a pressure that could gradually stabilise ordering preferences across repeated language use (Franke, Scontras, and Simonic, 2019). Sequential context updating provides a mechanism for this advantage: the noun-adjacent adjective restricts the context in which an outer, more subjective adjective is interpreted. The present production findings place this proposal within a broader set of choices, because speakers also decide which adjectives to include and how much to say. The theoretical question is therefore how the proposed ordering advantage changes when the speaker evaluates information available at each adjective position and can choose descriptions of different lengths.

The simulation addresses this question through controlled changes on the same randomly generated reference scenes. It begins with a comparison of the two size–colour orders based on their completed meanings, then introduces the utterance evaluation used by the production model and expands the alternatives to include single adjectives and the full production utterance space. Within that full utterance space, it then compares global, plan-guided, and fully incremental production under the same semantic settings. The scene distribution and listener target-recovery measure remain fixed throughout. This comparison identifies the conditions under which the advantage depends on semantic updating and those under which it also arises with fixed meanings. The simulation estimates the average direction and magnitude of the advantage as these production assumptions change across scenes.

6.1 Simulation Design and Size-First Advantage

The simulation uses 24,000 randomly generated displays that vary in the number of objects and their size distribution. The largest object is designated as the target and assigned the colour blue, ensuring that both size and colour apply. The number of blue distractors varies across displays, so colour restricts the comparison class to subsets of different sizes. This extends the analysis beyond the experiment's controlled sufficiency conditions and varies how strongly colour-based restriction can affect the interpretation of size. Greater size spread also tends to separate the largest target more clearly from its competitors, providing a simulation analogue of the experimental size-discriminability manipulation. The same displays are reused across all comparisons. The scene-generation procedure and semantic parameter grid appear in Appendix H.

The semantic regimes follow the distinction in Section 4: size is interpreted relative to the full scene under context-fixed semantics and relative to the context established by inner modifiers under sequential context updating. The first comparison uses *complete-utterance evaluation*: a global speaker chooses between *big blue* and *blue big* using only the literal listener's target probability after the completed utterance. The next comparison uses the *incremental evaluation* rule introduced in Section 4.2, accumulating listener log probabilities at each adjective position while retaining global choice ($\kappa = 0$). For *big blue*, this includes the information supplied by *big* as well as the completed description. Both comparisons use global choice; the difference is which information contributes to the utterance's utility. Holding the choice architecture fixed isolates the contribution of information available at each adjective position.

The global model is then given four alternatives by adding *big* and *blue*, followed by the fifteen one-, two-, and three-adjective utterances used in production. These changes allow the speaker to select a sufficient single adjective or include other properties. These four comparisons use global choice throughout and average over the same grid of 125 semantic parameter settings (Figure 10, Panel A). Panel B retains the fifteen utterances and uses the production semantic constants to compare global ($\kappa = 0$), plan-guided ($\kappa = .5$), and fully incremental ($\kappa = 1$) production. All three architectures use incremental evaluation; they differ in the contribution of local adjective competition. Pragmatic optimality remains $\alpha = 1$, with order, salience, and task-specific terms set to zero and final Laplace smoothing omitted. Further semantic settings and the full computational specification appear in Appendix H.1.

The pragmatic listener L^{prag} applies Bayes' rule to the speaker distribution: an object receives more probability when a speaker intending that object would be more likely to produce the observed utterance. All objects have equal prior probability. For each context, the size-first advantage compares the listener's target probability after the two orders:

$$\Delta_{\mathrm{SF}} = L^{\mathrm{prag}}(o^* \mid \text{“big blue”}) - L^{\mathrm{prag}}(o^* \mid \text{“blue big”}). \tag{8}$$

Positive values indicate that the pragmatic listener assigns a higher probability to the target after the size-first order. Because the listener reasons about the speaker's choice, other available descriptions can affect this measure even though the two utterances being evaluated remain the same.

6.2 Conditions under which an Order Asymmetry Emerges

With global choice and complete-utterance evaluation, sequential context updating generated a mean size-first advantage of approximately 1.1 percentage points, while context-fixed semantics yielded no difference between the orders (Figure 10, Panel A). With fixed meanings, the two adjective orders have the same completed interpretation and therefore the same target probability under this evaluation rule. Updating breaks this symmetry because colour restricts the comparison class for size. This comparison recovers the proposed communicative contribution of hierarchical restriction.

Including information available at each adjective position changed the result. With only the two orders available, the global production model yielded a mean colour-first advantage under both semantic regimes: approximately 2.7 percentage points with fixed meanings and 1.8 with updating. The first adjectives can differ in informativeness even when the completed meanings coincide, allowing an ordering asymmetry under fixed semantics. Its direction, however, depends on the alternatives available for reference. Adding the single-adjective responses changed the mean advantage to size-first under both regimes, to approximately 2.3 and 2.8 percentage points, respectively. The speaker can now identify a referent with one adjective, changing how probability is distributed over the two-adjective descriptions and how the pragmatic listener interprets their use. Adjective selection and utterance length thus affect the communicative value of an order even when the two orders themselves remain unchanged.

Expanding to fifteen utterances retained the mean size-first advantage, at approximately 2.6 percentage points with fixed semantics and 3.1 with updating. At the semantic constants used in production, all three production architectures generated a mean size-first advantage under both semantic regimes (Figure 10, Panel B). The advantage was approximately 1.9 percentage points for global production, 1.5 for plan-guided production, and 1.0 for fully incremental production, with nearly identical values for fixed and updating semantics within each architecture. Increasing the contribution of local adjective competition reduced the magnitude of the advantage while preserving its direction in this comparison. Updating's additional contribution therefore depends on the semantic settings, and the size-first advantage in the broader utterance space extends across the three production architectures.

The simulation shows how the interpretation and selection of adjectives jointly determine the communicative advantage of an order. Semantic updating can generate an advantage when comparing completed meanings; evaluating information available at each adjective position and allowing shorter descriptions changes both the source and direction of that advantage. The effect of adding single-adjective alternatives gives a computational basis for considering order within the broader production distribution: the option of using a shorter description affects how the listener interprets the same two-adjective utterances.

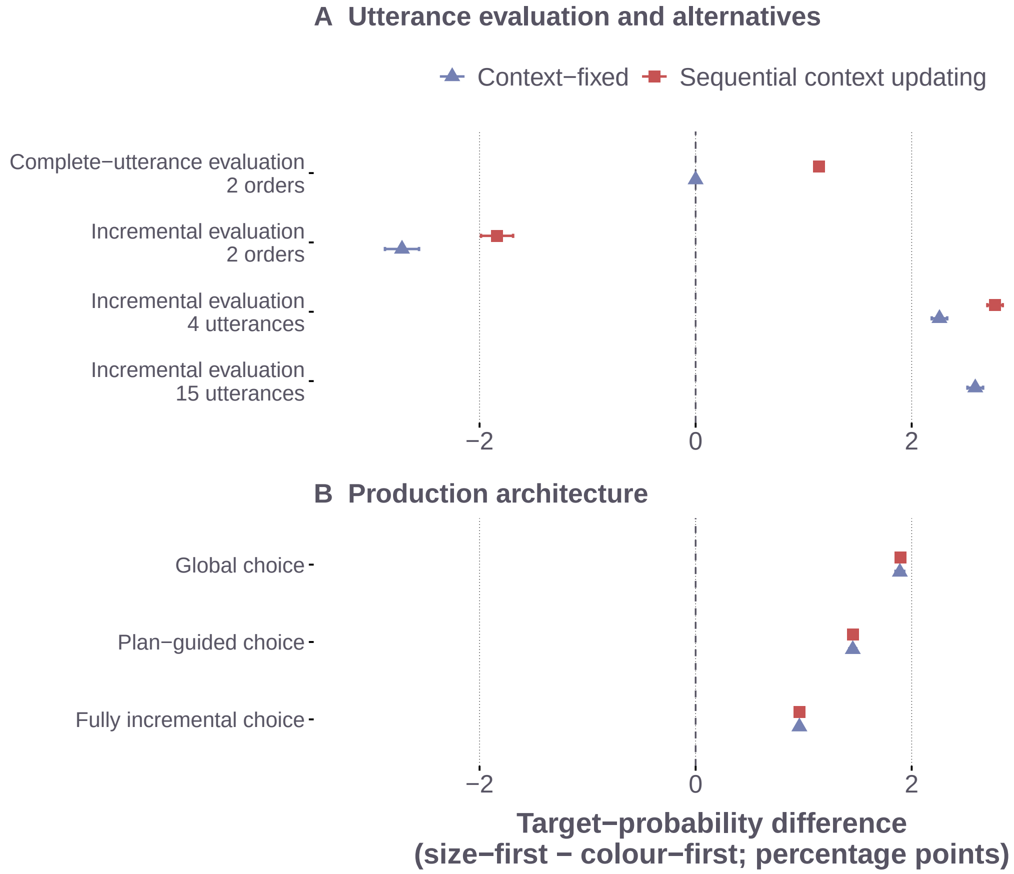


Figure 10: Size-first pragmatic target-recovery advantage across utterance evaluation, alternatives, and production architecture on the same 24,000 random scenes. Panel A holds global choice fixed and averages over the original semantic parameter grid. Complete-utterance evaluation uses only the final literal-listener target probability; incremental evaluation sums listener log probabilities at each adjective position. The four-utterance space adds single adjectives to the two orders; the fifteen-utterance space contains all production responses. Panel B holds the fifteen utterances and production semantic constants fixed while comparing global ($\kappa = 0$), plan-guided ($\kappa = .5$), and fully incremental ($\kappa = 1$) production. All three use incremental evaluation. Points show scene-averaged target-probability differences; positive values favour size-first and negative values favour colour-first. Intervals are ± 1.96 Monte Carlo standard errors, calculated after averaging parameter settings within scenes.

7 General Discussion

The present findings connect adjective ordering to the cognitive organisation of reference production. Stable order preferences coexist with context-sensitive choices of descriptive content, and plan-guided incremental production provides the best predictions of the resulting utterances. This combination raises broader questions about how speakers construct useful descriptions under limited planning resources, maintain semantic representations during production, and determine which alternatives to consider. These questions concern the processes through which communicative goals give rise to adjective selection and order.

7.1 Computational Intractability and Bounded Planning

A communicatively useful description must be selected from alternatives that differ in content, length, and order. As the number of available properties increases, the number of possible combinations and orderings grows combinatorially, making exhaustive comparison increasingly demanding. The cognitive problem is therefore how speakers preserve sensitivity to the communicative consequences of a description while limiting the alternatives and continuations they evaluate. Computational tractability places constraints on the processes and representations that can support such behaviour (Rooij, 2008). The relevant demands depend on how the problem is structured: restricting the candidate descriptions or the extent of prospective evaluation can change the computation required.

Approximation alone provides no general guarantee of tractability (Kwisthout, Wareham, and Van Rooij, 2011).

Plan-guided production motivates a hypothesis about how planning could be organised under these constraints. An utterance-level plan could specify properties relevant to identifying the target, constraining the alternatives considered during incremental word choice. Local choices could then remain sensitive to the developing description and the properties still available for mention. This organisation would allow speakers to anticipate the contribution of later information while limiting the scope of each decision. It accords with evidence that conceptual planning and linguistic formulation can proceed with different scopes during production (Griffin and Bock, 2000; Ferreira and Swets, 2002; Konopka and Meyer, 2014).

The present model specifies a probability distribution over completed utterances and evaluates the full response space. Developing this account at the process level requires specifying how speakers generate candidate descriptions and allocate computation to their evaluation. A resource-rational approach would relate the benefit of further planning to its time and processing costs (Lieder and Griffiths, 2020). One concrete possibility is that speakers consider a limited set of prospective descriptions and revise it as words are selected. Manipulating planning time and the number of relevant properties would then test whether additional planning changes earlier choices through the information available in later adjectives. Such a test would connect the model's joint sensitivity to utterance-level utility and local competition to the resources used during production.

7.2 Semantic Stability during Incremental Production

Incremental reference production requires coordination between a representation of the visual scene and the interpretation of the developing utterance. The context-fixed model illustrates how a size comparison established over the scene can remain available as additional adjectives narrow the listener's candidate set. Its predictive advantage makes the maintenance of a scene-based comparison class a plausible component of this coordination. A speaker could establish which objects count as large or small during scene apprehension and use that representation while selecting a description. The stability of this comparison would provide continuity across choices involving different properties and utterance lengths.

This interpretation raises a question about when a comparison class is established and when it is revised. The full scene and marked target in the present task provide information from which a size comparison can be formed before linguistic production begins. In discourse, a subset of referents may instead become relevant through the preceding utterance, potentially changing the comparison against which size is evaluated. Varying the timing of contextual information would help identify the conditions under which speakers retain an initial comparison or establish a new one. Independently varying lexical subjectivity and visual discriminability would further distinguish uncertainty about a property's interpretation from difficulty perceiving that property in a particular scene. These extensions would connect the comparison of semantic regimes to a broader account of how perceptual and linguistic context guide the construction and maintenance of meaning.

7.3 Alternative Generation and Communicative Efficiency

The availability of a shorter description changes how a pragmatic listener interprets the use of a longer one. The simulation demonstrates the consequence for adjective order: introducing single-adjective alternatives changes the average communicative advantage of the same two-adjective sequences. This dependence makes the set of alternatives a substantive part of a cognitive account. An ordering preference can depend on which descriptions the speaker considers and which alternatives the listener expects the speaker to have considered. The communicative value of an order consequently depends on the relation between these perspectives.

The next explanatory question concerns how alternatives become available during reference production. Visual attention, lexical accessibility, and the current discourse may influence which properties enter consideration and how they are combined into candidate descriptions. An utterance-level plan could constrain this process by directing consideration towards properties relevant to the referential goal, while incremental choice determines which of those properties is mentioned next. On this account, the planning problem includes both constructing a set of useful candidates and choosing among them. A process model could represent this distinction by generating candidate descriptions before evaluating their communicative utility.

This proposal yields a test beyond changing the discriminatory strength of the adjectives themselves. Making shorter alternatives more or less available, while holding the scene and the two-adjective sequences constant, should affect their interpretation if listeners take those alternatives into account. Comparing production and comprehension under the same manipulation would examine whether the alternatives speakers consider correspond to those listeners use in pragmatic reasoning. Such evidence would extend the present link between adjective selection and ordering to the coordination of speaker and listener expectations.

7.4 Redundancy and Adaptive Planning

Redundant descriptions offer a useful case for examining how production processes and communicative goals interact. A property may be selected because it becomes accessible early, because it helps the listener identify the target, or because both pressures favour its inclusion (Pechmann, 1989; Engelhardt, Bailey, and Ferreira, 2006; Rubio-Fernández, 2016; Degen et al., 2020). In a shared visual scene, readily encoded information may also be useful for visual search or uncertainty reduction. The cognitive explanation therefore depends on how accessibility and anticipated listener benefit jointly affect the decision to include another adjective.

The improvement from participant-specific incremental-choice weights suggests that the balance represented by the model varies across speakers. A further possibility is that this balance adapts within a speaker to the demands of the referential situation. Additional planning may be worthwhile when a description is difficult to construct or misunderstanding is costly; accessible properties may exert a greater influence when preparation time is limited. This is a resource-rational hypothesis about the allocation of planning effort (Lieder and Griffiths, 2020). It predicts systematic changes with task demands that can be distinguished from stable participant differences by manipulating planning opportunity within speakers.

Independently manipulating property accessibility and listener uncertainty would clarify how these influences contribute to redundancy. Speech onset and eye movements could establish whether an additional property was available before production began or entered consideration during the utterance. The resulting evidence would relate description length to the time course of planning and reveal when additional information serves the listener's needs. Freer production with new participants and materials would further test these relationships beyond the fixed vocabulary and German prenominal sequences examined here.

The broader contribution is an account of adjective ordering embedded in the cognitive problem of constructing a useful referential description. Stable order expectations, scene-based meanings, and incremental competition provide different resources for this process. The plan-guided framework connects their contribution to the content and length of the utterance, while the simulations show how the available alternatives affect its interpretation. Explaining how these resources are coordinated under the demands of communication links adjective order to what speakers choose to mention, how much they choose to say, and how they plan to say it.

8 Data Availability Statement

Data, analysis scripts, model code, posterior summaries, and figure-generation scripts will be made available in an anonymised repository upon submission. The repository will include the production-model inputs and outputs, documentation of preprocessing and input reconstruction, and the scripts required to reproduce the reported behavioural and predictive analyses.

9 Author Note

Acknowledgements, funding information, author contributions, and competing-interest statements will be finalised with the author team before journal submission.

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A Supplementary Experimental Methods and Results

Table 1 summarises the participant exclusions and the observations included in each analysis. Participants could meet more than one exclusion criterion; the table therefore reports both criterion-specific counts and the total number excluded. For Study 2, the production model comparison reported in Section 5 uses all 9,100 annotated observations across the fifteen utterance categories and three adjective-pair families. The behavioural analyses focus on the size–colour subset.

Table 1: Participant exclusions and observations included in each analysis. Criterion counts overlap within study.

Experiment	Starting and final N	Participant exclusions	Observations included in the manuscript
Study 1a slider ratings	$150 \rightarrow 134$	8 with more than three missing trial-list entries; 8 with extreme completion time; 2 with more than three control-item failures; 6 with more than three "none fits" responses on experimental items; 16 unique participants excluded.	All retained critical trials: 10,606 observations. Reported size–colour slider subset: 3,485 observations.
Study 1b slider replication	$132 \rightarrow 104$	6 missing-list failures; 1 duration failure; 3 control failures; 24 experimental none-fit failures; 28 unique participants excluded.	All retained replication trials: 8,379 critical observations. Reported size–colour slider subset: 2,786 observations.
Production study	$149 \rightarrow 113$	8 with extreme completion time; 25 with more than 80% identical responses; 7 with more than 10% missing or unannotatable responses; 36 unique participants excluded.	Full production dataset: 9,100 annotated observations. Reported size–colour behavioural subset: 3,034 observations.

The slider-rating materials used the fixed adjective vocabulary in Table 2. The raw stimulus table encodes colours and forms with compact feature labels; the interface displayed the corresponding German adjective forms in phrases such as *den großen blauen Aufkleber*.

Table 2: Adjective vocabulary in the slider-rating item descriptions.

Property class	Raw stimulus values	German adjective forms displayed in the item text
Size	Numeric size levels `1--6`, `9`, `10`	*großen*, *kleinen*
Colour	`blue`, `orange`, `grey`, `brown`, `green`, `black`	*blauen*, *orangen*, *grauen*, *braunen*, *grünen*, *schwarzen*
Form	`circle`, `diamond`, `heart`, `quadrat`, `star`, `triangle`	*runden*, *diamantförmigen*, *herzförmigen*, *quadratischen*, *sternförmigen*, *dreieckigen*

For modelling, each response is matched to its recorded display. Colour and form are represented by whether each object matches the target on that property; the lexical forms shown to participants are listed above.

The first author annotated production responses using the 15 ordered adjective categories listed in Table 3. The data files use `D` for the size or dimension adjective. The model exposition uses S; the two notations refer to the same adjective class. Misspellings and transparent lexical alternatives were normalised when their semantic class was unambiguous. Spatial descriptions, comparative or ordinal size utterances, and responses that could not be assigned to a target adjective class were excluded from the production model comparison.

Table 3: Production annotation scheme. In the data files, `D` corresponds to the size adjective S; `C` and `F` correspond to colour and form.

Length	Codes
One adjective	`D`, `C`, `F`
Two adjectives	`DC`, `DF`, `CD`, `CF`, `FD`, `FC`
Three adjectives	`DCF`, `DFC`, `CDF`, `CFD`, `FDC`, `FCD`

A.1 Behavioural Analysis Specification

The behavioural hypotheses were evaluated with Bayesian hierarchical models. All analyses used the size–colour subsets reported in the main experiment figures: 3,485 observations in Study 1a, 2,786 in Study 1b, and 3,034 in Study 2. The later production-model comparison addresses predictive differences among the proposed computational mechanisms over the complete fifteen-category response space and all 9,100 production trials. For the slider analyses, ratings were reoriented toward size-first, divided by 100, and centred by subtracting 0.5. Let $\mathbf{x}_{ij}$ encode referential context, size discriminability, and their interaction, and let $\mathbf{z}_{ij}$ encode the participant intercept and context slopes. The slider models used a Gaussian likelihood with

$$y_{ij} \sim \mathrm{Normal}(\mu_{ij}, \sigma), \qquad \mu_{ij} = \mathbf{x}_{ij}^{\mathsf{T}}\boldsymbol{\beta} + \mathbf{z}_{ij}^{\mathsf{T}}\mathbf{b}_i, \qquad \mathbf{b}_i \sim \mathrm{MVN}(\mathbf{0}, \boldsymbol{\Sigma}_{\mathrm{participant}}).$$

For production, size-initiality indicated whether the coded response began with the size adjective. Redundant adjective use, coded as over-informativeness in the analysis, indicated a three-adjective response in both-necessary contexts or a multi-adjective response in one-property-sufficient contexts. Both binary outcomes used a Bernoulli likelihood with a logit link and an item intercept c_k:

$$y_{ijk} \sim \mathrm{Bernoulli}(\pi_{ijk}), \qquad \mathrm{logit}(\pi_{ijk}) = \mathbf{x}_{ijk}^{\mathsf{T}}\boldsymbol{\beta} + \mathbf{z}_{ijk}^{\mathsf{T}}\mathbf{b}_i + c_k, \qquad c_k \sim \mathrm{Normal}(0, \sigma_{\mathrm{item}}).$$

For the slider studies, the interaction was represented by two posterior contrasts: the high-minus-low discriminability effect in the both-necessary context minus the corresponding effect in the size-sufficient context, and the high-minus-low effect in the colour-sufficient context minus the effect in the size-sufficient context. The production interactions were planned contrasts on the posterior-predictive probability scale. For size-initiality, the contrast was the high-minus-low discriminability effect in the both-necessary context minus the mean effect in the two one-property-sufficient contexts. For redundant adjective use, the contrast was the low-minus-high discriminability effect in the size-sufficient context minus the mean effect in the both-necessary and colour-sufficient contexts.

The slider models used a Gaussian likelihood, correlated participant intercepts and context slopes, fixed-effect priors $\mathrm{Normal}(0, .30)$, $\mathrm{HalfNormal}(.20)$ priors on participant standard deviations, and a $\mathrm{HalfNormal}(.25)$ prior on residual standard deviation. The production models used Bernoulli likelihoods, the same correlated participant effects, an item intercept, a $\mathrm{Normal}(0, 2)$ intercept prior, $\mathrm{Normal}(0, 1.5)$ priors on the remaining fixed effects, and $\mathrm{HalfNormal}(1)$ priors on participant and item standard deviations. Participant-effect correlations received an $\mathrm{LKJ}(2)$ prior. Four chains run in parallel each provided $1{,}000$ retained draws after $1{,}000$ warm-up iterations. All four fits had zero divergences, no maximum-tree-depth hits, $\widehat{R} \le 1.01$, and minimum effective sample sizes above 900. For each posterior draw, posterior-predictive cell expectations were computed for a new participant and, in production, a new item by integrating over the fitted random-effects distributions. The planned contrasts were then evaluated on the rating or response-probability scale.

A.2 Additional stimulus and procedure details

Critical scenes were realised through systematic permutations of colour and form values applied to 27 six-object scene templates. This varied the surface displays while preserving the referential structure of each condition.

The slider stimulus inventory contains six lists of 180 main trials: 81 critical trials and 99 fillers and controls per list, plus three practice displays. Lists 1–3 use high size discriminability and lists 4–6 use low size discriminability. Each list contains nine critical trials for each combination of the three adjective families and three referential conditions. The large-size band comprises $\{9, 10\}$ in both groups; the smaller-size band is $\{1, 2, 3\}$ at high discriminability and $\{1, 2, 3, 4, 5, 6\}$ at low discriminability. Thus, contrasts between the large and smaller bands can span 6–9 recorded units in the high-discriminability materials and 3–9 units in the low-discriminability materials. The stimulus table specifies the six object sizes, colours, and forms for every list–item–condition combination.

Each study began with three practice trials with feedback. Main trials were divided into four blocks of approximately equal length, with self-paced breaks and independently randomised trial order within each block. Study 1a participants completed 81 critical trials and 99 filler or control trials (180 total), with a mean completion time of 27.6 minutes. Control trials had an unambiguous expected response; the additional "none fits" option allowed participants to indicate that neither description identified the target. Frequent use of this option on experimental trials was an exclusion criterion, as detailed in Table 1. Study 1b used the same 1,080 combinations of list, item, and condition as Study 1a.

Study 2 participants completed 81 critical and 54 filler or control trials (135 total), with a mean completion time of 32.4 minutes. Fewer filler and control trials were used because typed responses were slower and more effortful than slider judgements. Compensation followed the same rate as Study 1a. Restricting responses to the adjective slot reduced off-task variability and provided a controlled basis for annotation. The categorical response variable for the computational comparison is the ordered utterance label $y_i \in \{1, \ldots, 15\}$. For each binary behavioural outcome, matching responses were coded as 1 and all other annotated size–colour responses as 0.

A.3 Detailed behavioural results

Tables 4 and 5 report posterior medians and 95% credible intervals for the contrasts described in the main text. Slider estimates are in centred-rating units; production estimates are differences in response probability. The column P reports the posterior probability of the direction stated in each row; a dash indicates that a directional probability was not separately reported for that contrast in the main analysis summary. Interaction estimates compare effects across contexts, whereas within-context contrasts compare the two discriminability groups in one context.

Table 4: Slider-rating contrasts, with interactions followed by context contrasts and within-context discriminability effects. Discriminability effects are high minus low. Interaction rows subtract the size-sufficient discriminability effect from the effect in the named context.

Study	Contrast	Median	95% CrI	P (direction)
1a	Interaction: both-necessary	$-.085$	$[-.151, -.022]$	.995 (< 0)
1a	Interaction: colour-sufficient	$-.063$	$[-.172, .039]$	.885 (< 0)
1a	Size-sufficient minus both-necessary	.094	$[.061, .127]$	—
1a	Both-necessary minus colour-sufficient	.171	$[.131, .208]$	—
1a	Colour-sufficient rating relative to zero	.054	$[.003, .109]$	.982 (> 0)
1a	Discriminability: size-sufficient	.042	$[-.013, .099]$	—
1a	Discriminability: both-necessary	$-.043$	$[-.118, .028]$	—
1a	Discriminability: colour-sufficient	$-.021$	$[-.129, .082]$	—
1b	Interaction: both-necessary	$-.014$	$[-.097, .071]$	.623 (< 0)
1b	Interaction: colour-sufficient	.034	$[-.082, .157]$	.285 (< 0)
1b	Size-sufficient minus both-necessary	.073	$[.029, .115]$	—
1b	Both-necessary minus colour-sufficient	.140	$[.090, .190]$	—
1b	Colour-sufficient rating relative to zero	.057	$[.001, .116]$	.977 (> 0)
1b	Discriminability: size-sufficient	.090	$[.014, .160]$	—
1b	Discriminability: both-necessary	.076	$[-.011, .161]$	—
1b	Discriminability: colour-sufficient	.124	$[.016, .240]$	—

The posterior probability of the graded context order was greater than .999 in Study 1a and .999 in Study 1b. Observed mean centred ratings in Study 1b were .272 in size-sufficient, .197 in both-necessary, and .054 in colour-sufficient contexts. Figure 11 shows the observed Study 1a ratings; the main-text figure shows fitted cell expectations with 95% credible intervals for both slider studies.

Observed redundancy proportions in size-sufficient displays were .90 under low and .63 under high size discriminability. Observed size-initial proportions in both-necessary displays were .49 under low and .65 under high discriminability. These descriptive proportions differ from the posterior-predictive contrasts in Table 5, which integrate over participant and item variation.

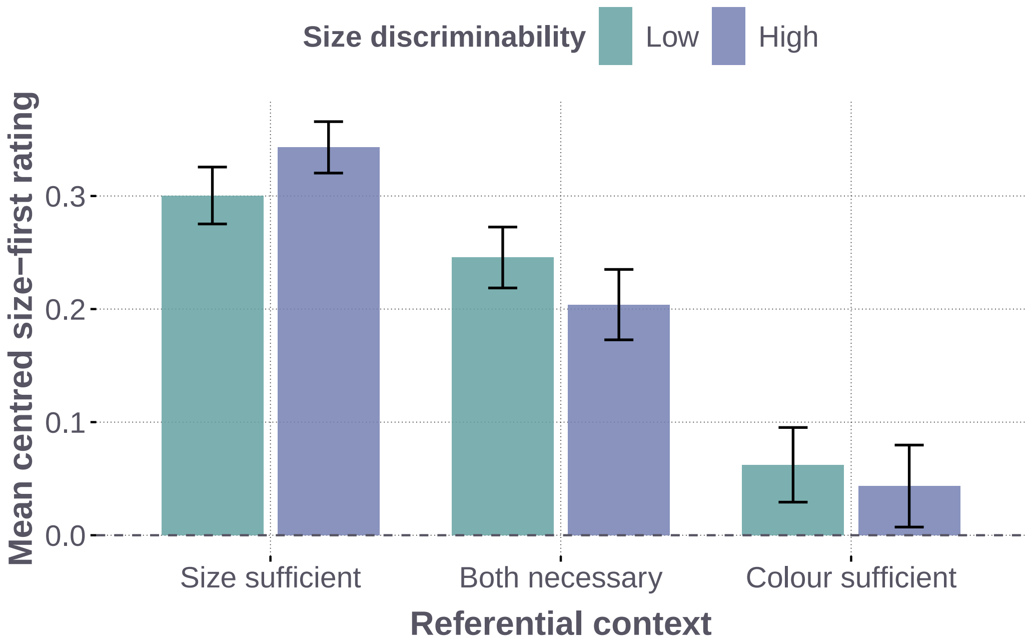

Figure 11: Observed Study 1a slider ratings. Mean centred trial ratings by referential context and size discriminability, with descriptive 95% t intervals; positive values indicate size-first preference.

Table 5: Production contrasts on the response-probability scale. The redundancy interaction compares the low-minus-high difference in size-sufficient displays with its mean in the other two contexts. The size-initiality interaction compares the high-minus-low difference in both-necessary displays with its mean in the two one-property-sufficient contexts.

Contrast	Median	95% CrI	P (direction)
Redundancy: planned interaction	.153	[.072, .236]	$>$ .999 ($>$ 0)
Redundancy: size-sufficient, low minus high	.234	[.151, .318]	$>$ .999 ($>$ 0)
Size-initiality: planned interaction	.169	[.077, .256]	$>$ .999 ($>$ 0)
Size-initiality: both-necessary, high minus low	.141	[.036, .242]	.997 ($>$ 0)
Size-initiality: size-sufficient, high minus low	−.021	[−.101, .057]	—
Size-initiality: colour-sufficient, high minus low	−.032	[−.068, −.006]	—
Size-initiality: size-sufficient minus both-necessary	.193	[.139, .248]	—
Size-initiality: both-necessary minus colour-sufficient	.531	[.459, .601]	—

A descriptive check of the complete production data also showed group differences in the colour–form family. In form-sufficient displays, form-initial responses were more frequent in the high- than in the low-discriminability group (.77 vs. .71). Because discriminability was manipulated between participants, these differences can also reflect group variation in how the task was approached. The production model represents participant variation alongside the visual manipulation.

B Production Model Specification

B.1 Representation of Experimental Displays

Participant list, condition, and item identifiers link each retained response to its recorded six-object display. The model represents each display as a 6×3 array of size, colour, and form values, with the intended referent indexed first. This representation preserves the observed values of all three properties, including those outside the adjective pair defining the trial.

B.2 Graded Size Meaning

Size is represented by a graded threshold over the comparison class $\mathcal{C}$ (Schmidt et al., 2009; Franke, Scontras, and Simonic, 2019):

$$\theta_\delta(\mathcal{C}) = Q_{.80}(\mathcal{C}) - \delta\left(Q_{.80}(\mathcal{C}) - Q_{.20}(\mathcal{C})\right), \tag{9}$$

where $Q_{.20}$ and $Q_{.80}$ are probability-weighted size quantiles and δ controls threshold selectivity. For an object of size s_o, graded membership is

$$m_{\mathrm{S}}(o \mid \mathcal{C}) = \Phi\left(\frac{s_o - \theta_\delta(\mathcal{C})}{w\sqrt{s_o^2 + \theta_\delta(\mathcal{C})^2 + \epsilon}}\right), \qquad w = .6856, \quad \epsilon = 10^{-8}, \tag{10}$$

where Φ is the standard normal cumulative distribution function. The weighted quantiles are the first observed sizes at which cumulative comparison-class probability reaches .20 and .80. The same width w is used in both discriminability groups; the recorded sizes enter this function directly. For colour and form, matching objects receive membership ν and nonmatching objects $1-\nu$, with $\nu_{\mathrm{C}} = .59$ and $\nu_{\mathrm{F}} = .50$. Each prefix is interpreted from the uniform object prior, applying its modifiers once in noun-outward order. Context updating changes the comparison-class weights used for the size quantiles as inner modifiers are applied. Equations (1), (2), (4) and (5) define how this meaning enters the literal listener L and the three production accounts.

B.3 Component Inventory

Table 6: Components of the production models.

Component	Status	Role
Observed display input	Fixed	All visible size, colour, and form values
Size threshold and blur	Fixed constants	Graded size meaning in the scene comparison class
Colour reliability	Fixed	Graded support for matching colour
Form reliability	Fixed neutral	Form production supported by the task link
Pragmatic rationality	Estimated, hierarchical	Participant variation in response concentration
Stable-order score Ω	Fixed feature	Relative order expectation within adjective sets
Stable-order weight β_i	Estimated, hierarchical	Participant variation in stable order expectation
Salience coefficient	Estimated	Display prominence in candidate utility
Task-link coefficients	Estimated	Support for sufficient single adjectives, form use, and length and discriminability effects
Incremental-choice weight κ_i	Estimated in plan-guided production	Shared weight or participant-specific weights for incremental adjective choice

The population pragmatic-optimality parameter has prior $\alpha_0 \sim \mathrm{HalfNormal}(3)$, and the population log stable-order weight has prior $\mu_{\log\beta} \sim \mathrm{Normal}(0, .35)$. Participant effects are noncentred: $\alpha_i = \alpha_0 \exp(\tau_{\log\alpha} z_{\alpha i})$ and $\beta_i = \exp(\mu_{\log\beta} + \tau_\beta z_{\beta i})$, with independent standard-normal z values, $\tau_{\log\alpha} \sim \mathrm{HalfNormal}(.15)$, and $\tau_\beta \sim \mathrm{HalfNormal}(.5)$. The shared plan-guided weight has prior $\kappa \sim \mathrm{Uniform}(0,1)$. In the model with participant-specific incremental-choice weights, $\mu_\kappa \sim \mathrm{Uniform}(0,1)$ and $\kappa_i = \mathrm{logit}^{-1}(\mathrm{logit}(\mu_\kappa) + \tau_\kappa z_{\kappa i})$, with $\tau_\kappa \sim \mathrm{HalfNormal}(.5)$ and independent $z_{\kappa i} \sim \mathrm{Normal}(0,1)$. The salience coefficient has a $\mathrm{HalfNormal}(1)$ prior and the salience-continuation coefficient a $\mathrm{HalfNormal}(.75)$ prior. The sufficient-single-adjective and form-use coefficients each have a $\mathrm{HalfNormal}(1.5)$ prior; the three-adjective penalty and the two size-reliability coefficients each have a $\mathrm{HalfNormal}(1)$ prior. Threshold selectivity is fixed at $\delta = .50$; colour reliability, form reliability, and perceptual blur take the values specified above. A final Laplace smoothing step mixes the predicted distribution with a uniform distribution over the fifteen responses, with a fixed weight of .003 on the uniform component.

Visual salience. The salience feature combines a fixed preference across properties with their contrast in the displayed scene. For the target o^*, let s_o, c_o, and f_o be an object's recorded size, colour, and form, and let h indicate high size discriminability. The three contrast scores are

$$r_{\mathrm{S}} = \frac{1+h}{2}\,\mathrm{logistic}\left(\frac{s_{o^*} - \max_{o \neq o^*} s_o}{\mathrm{sd}(s) + 10^{-8}}\right), \tag{11}$$

$$r_{\mathrm{C}} = 1 - \frac{1}{5}\sum_{o \neq o^*} \mathbf{1}\{c_o = c_{o^*}\}, \qquad r_{\mathrm{F}} = 1 - \frac{1}{5}\sum_{o \neq o^*} \mathbf{1}\{f_o = f_{o^*}\}. \tag{12}$$

Here $\mathrm{sd}(s)$ is the population standard deviation across the six displayed sizes. The fixed baseline scores are $(b_{\mathrm{S}}, b_{\mathrm{C}}, b_{\mathrm{F}}) = (0, 1, .25)$. Centring across the three properties gives

$$z_a = b_a + r_a - \frac{1}{3}\sum_v (b_v + r_v).$$

A property receives greater salience when it contrasts with the distractors; the size contrast is additionally scaled by discriminability. The same feature enters prefix utility through $\lambda_{\mathrm{salience}} z_a$ and the continuation cost in the task link below.

Stable order expectations. We derived the fixed order feature from `dbmdz/german-gpt2`. For each pairing of six colour and six form adjectives, the scoring script generated all fifteen non-empty orders using those adjectives and *großer*, followed by the noun *Aufkleber*, without a surrounding sentence. It evaluated the resulting 540 strings, including repeated shorter strings, using mean token surprisal in bits. We averaged this surprisal within each order category and normalised $2^{-\bar{s}(u)}$ across the fifteen categories to obtain the fixed weights $q(u)$. For $\mathcal{U}_{A(u)}$, the set of orders expressing the same adjective set as u, the model uses

$$\Omega(u) = \log q(u) - \frac{1}{|\mathcal{U}_{A(u)}|}\sum_{v \in \mathcal{U}_{A(u)}} \log q(v). \tag{13}$$

This centring preserves relative order preferences within an adjective set and gives every single-adjective response a score of zero. The scoring script, archived token-surprisal table, aggregation code, and fixed scores and residuals are supplied with the model code. Participant-specific β_i determines the influence of these stable expectations when the model normalises across the complete response space.

Task link for adjective selection and length. The task link $T(u, X)$ connects the referential design to choices about which properties to mention and how far to continue a description. Let $n = |u|$ and $A(u)$ denote the length and unordered adjective set of u. For this term, X includes the display's condition labels: $d \in \{\mathrm{S}, \mathrm{C}, \mathrm{F}, \varnothing\}$ is the property that identifies the target on its own, with $d = \varnothing$ in both-necessary conditions; c indicates a colour-sufficient condition; and $h = 1$ indicates high size discriminability ($h = 0$ for low). Using $\mathbf{1}\{\cdot\}$ for an indicator, the production models use

$$\begin{aligned} T(u, X) = \; & -\rho_{\mathrm{cont}} \sum_{t=1}^{n-1} \max(z_{w_t}, 0) + \lambda_{\mathrm{suff}} \mathbf{1}\{d \neq \varnothing,\ u = (d)\} \\ & + \lambda_{\mathrm{F}}(1-c)\mathbf{1}\{\mathrm{F} \in A(u)\} - \lambda_3 \mathbf{1}\{n = 3\} \\ & + \lambda_{\mathrm{S}} h\, \mathbf{1}\{d = \mathrm{S},\ u = (\mathrm{S})\} \\ & + \lambda_{\mathrm{SF}}(1-2h)\mathbf{1}\{d = \mathrm{S},\ A(u) = \{\mathrm{S}, \mathrm{F}\}\}. \end{aligned} \tag{14}$$

The continuation cost sums the positive salience scores of adjectives followed by another adjective, using the same z scores as the prefix utility in Equation (2). The sufficiency term favours a single adjective that identifies the target, while the form term increases the value of form-containing descriptions outside colour-sufficient conditions. With form reliability fixed at .50, form gives equal literal support to matching and nonmatching objects; this task term provides a direct contribution to form use. The three-adjective penalty offsets the value accumulated along longer descriptions. In size-sufficient conditions, the final two terms additionally favour the single size adjective at high discriminability and favour size–form pairs at low discriminability while penalising them at high discriminability. The pair term applies equally to both orders. All six coefficients are nonnegative and shared across participants within each fitted model, with the priors given above. These terms enter the completed-utterance score alongside accumulated prefix utility, local competition, and the stable-order score; normalisation over all fifteen responses turns their combined values into predictions of adjective selection, length, and order.

Numerical evaluation and response likelihood. The implementation uses 64-bit arithmetic and a floor of 10^{-8} for size memberships, listener target probabilities before taking logarithms, and local choice probabilities. Colour and form memberships receive an additive 10^{-8} offset. Writing $g(u)$ for accumulated prefix utility, $i(u)$ for the sum of floored local log-choice probabilities, and $B(u) = \beta_i\Omega(u) + T(u, X)$, the computed score before Laplace smoothing is $(1 - \kappa_i)g(u) + \kappa_i i(u) + B(u)$. Normalising these scores yields $\widetilde{S}_{\kappa_i}(u \mid X)$, whose endpoints obey the geometric interpolation in Equation (6). The categorical response likelihood is

$$p(y_{it} = u \mid X_{it}) = (1 - .003)\widetilde{S}_{\kappa_i}(u \mid X_{it}) + \frac{.003}{15}. \tag{15}$$

The matched global and fully incremental models fix κ at zero and one, respectively; the matched plan-guided model estimates a shared κ; the model with participant-specific incremental-choice weights estimates κ_i. All models retain participant hierarchies for pragmatic optimality and stable-order weight, with the remaining task coefficients shared across participants.

C Production Model Comparison and Diagnostics

C.1 Inference and Predictive Comparison

Each likelihood evaluation scores all fifteen completed utterances and performs the required semantic composition and prefix-level normalisation. To make this exact calculation feasible, we precomputed trial-independent utterance and prefix structures and vectorised and compiled the listener and speaker computations in JAX. All eight production fits used four No-U-Turn Sampler (NUTS) chains on a GPU and 1,000 warm-up iterations per chain. The two global models retained 4,000 draws per chain at maximum tree depth 8 to obtain adequate mixing; the other six models retained 1,000 draws per chain at depth 6. The eight fits had zero divergences, maximum $\widehat{R} < 1.01$, minimum bulk and tail effective sample sizes above 400, minimum Bayesian fraction of missing information (BFMI) above .64, and no observation with Pareto $k > .7$. The participant-weight plan-guided fits reached the depth-6 step cap on 77.7% of transitions under fixed semantics and 31.7% under updating; these frequent depth-limit hits indicate restricted sampling efficiency, alongside the convergence and effective-sample-size values reported above. The best-predicting context-fixed model had an absolute ELPD of $-13{,}039.924$, maximum $\widehat{R} = 1.0051$, and minimum BFMI of .742.

Implementation checks verified that each modifier is applied once from the noun outwards and that changing the semantic regime can change predictions at matched parameters. Endpoint, interpolation, and direct-versus-precomputed probability checks agreed within numerical precision, and recomputed posterior log likelihoods agreed with the stored values within 10^{-8}.

The pointwise leave-one-out score vectors use the same trial ordering and response support. Because observations are repeated within participants, paired score contributions are first summed within participant. A participant-level Bayesian bootstrap then assigns Dirichlet weights to the 113 participant contributions over 200,000 draws. Reported uncertainty comprises 95% credible intervals and posterior probabilities for directional contrasts. The predictive target is another response from a participant represented in the dataset, as described in Section 5.1.

Table 7: Bayesian-bootstrap contrasts for the matched production architectures and the plan-guided participant hierarchy.

Contrast	ΔELPD	95% credible interval	$P(\Delta > 0)$
Plan-guided, shared κ, minus average endpoints	402.677	$[311.466, 503.098]$	$> .999$
Fully incremental minus global	-291.596	$[-684.673, -1.321]$	$.024$
Updating minus fixed, shared-κ grid	-43.527	$[-56.005, -31.770]$	$< .001$
Participant κ_i hierarchy, context-fixed	364.742	$[248.178, 520.967]$	$> .999$
Participant κ_i hierarchy, updating	372.285	$[255.597, 526.056]$	$> .999$

Averaging over the two semantic regimes, the plan-guided model had the highest predictive accuracy of the three architectures in .994 of the Bayesian-bootstrap draws. In the context-fixed plan-guided model, the posterior mean shared incremental-choice weight was

$$\kappa = .446, \qquad 95\% \text{ credible interval} = [.407, .486]. \tag{16}$$

C.2 Architecture-Level Response Patterns

Figure 12 shows initial-adjective cells in which the largest difference among architecture predictions is at least 6.7 percentage points and traces these differences into completed utterances. Colour-initial responses contributed 328.912 ELPD to the matched plan-guided model's advantage over the average endpoint, or 81.7% of the summed pointwise contrast. Across the displayed cells, plan-guided production often lies between the global and fully incremental endpoints. The response partitions locate this gain by summing exact-response log scores within the observed categories.

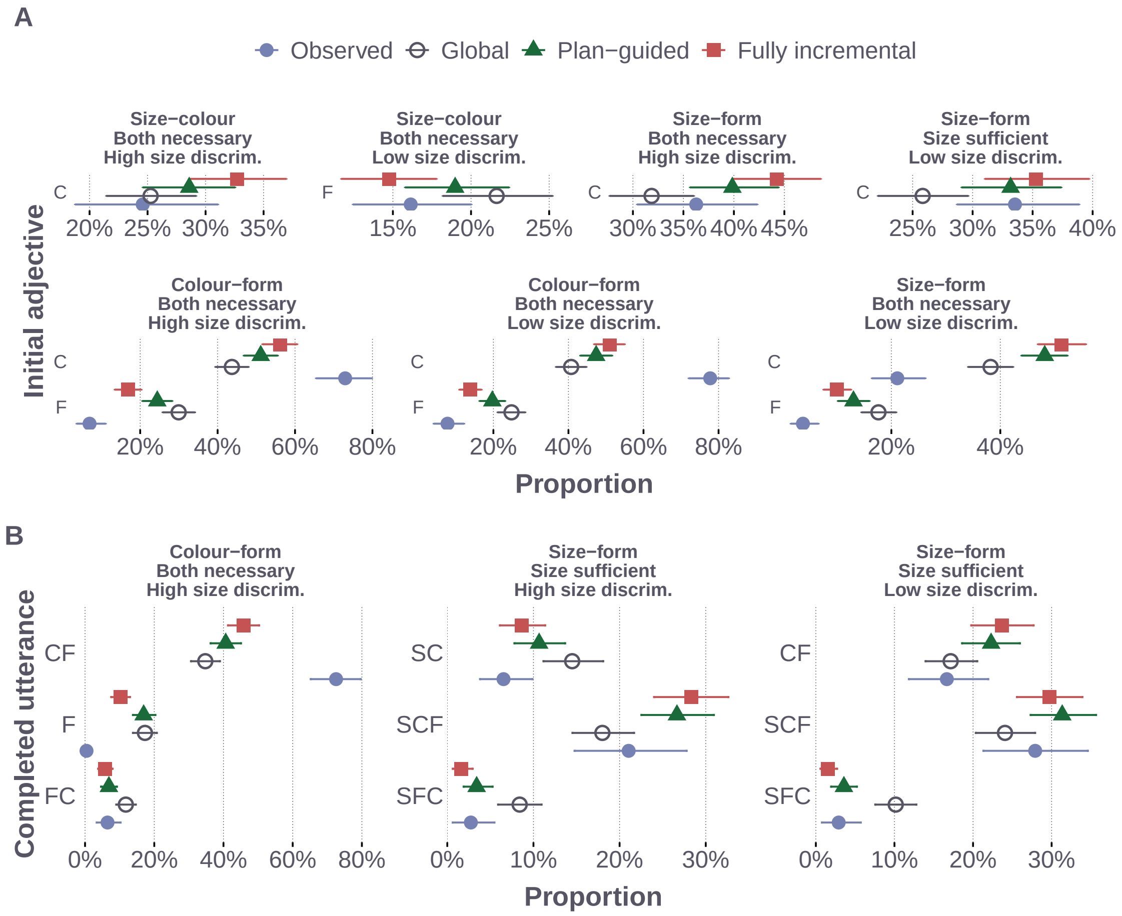

Figure 12: Qualitative comparison of the matched production architectures. Panel A shows initial-adjective cells for which the maximum difference among model predictions is at least 6.7 percentage points. Panel B shows completed utterances with differences of at least five percentage points within the three experimental conditions with the largest pairwise total-variation distance among model predictions. Response labels use S for size, C for colour, and F for form. Intervals show participant-bootstrap 95% intervals for observed proportions and 95% posterior-predictive intervals for model predictions. Horizontal scales vary across conditions to make the comparatively small differences among production architectures visible. The displayed cells are descriptive diagnostics; the held-out comparison in Figure 7 provides the evidence for relative predictive accuracy.

C.3 Decomposing the Semantic Predictive Contrast

The architecture-specific updating-minus-fixed effects were -45.104 ELPD for global production, -61.683 for plan-guided production, and -23.793 for fully incremental production. Each architecture-specific 95% credible interval excluded zero. The sequentially updating participant-specific model was lower by 54.140 ELPD, with a 95% credible interval for updating minus fixed of $[-69.042, -39.733]$.

Let $\ell(u)$ denote utterance length and $A(u)$ its unordered adjective set. For each held-out response, its predictive probability factors as

$$p_{-it}(u \mid X_{it}) = p_{-it}(\ell(u) \mid X_{it})\, p_{-it}(A(u) \mid \ell(u), X_{it})\, p_{-it}(u \mid A(u), X_{it}). \tag{17}$$

We first integrate the fifteen-response distribution using the PSIS weights for leaving out that entire response, then obtain the length and adjective-set probabilities by summation. Taking logarithms yields three additive contributions to the original response score. All three contributions refer to the same held-out response. Paired differences are summed within participant, with 95% intervals from 200,000 Dirichlet-bootstrap draws using seed 20260906. For the participant-weight plan-guided pair, updating minus fixed was -21.549 ELPD for length, 95% credible interval $[-30.933, -12.443]$, and -31.068 for adjective set conditional on length, $[-39.202, -23.530]$. The conditional-order contribution was -1.524 ELPD, with an interval of $[-4.879, 1.774]$. Across all four semantic pairs, adjective set conditional on length makes a negative contribution to updating minus fixed. For the participant-weight pair, the original response contrast sums to -17.548 ELPD in size–colour trials, -51.172 in size–form trials, and 14.580 in colour–form trials. These partitions locate the predictive difference within the complete response distribution.

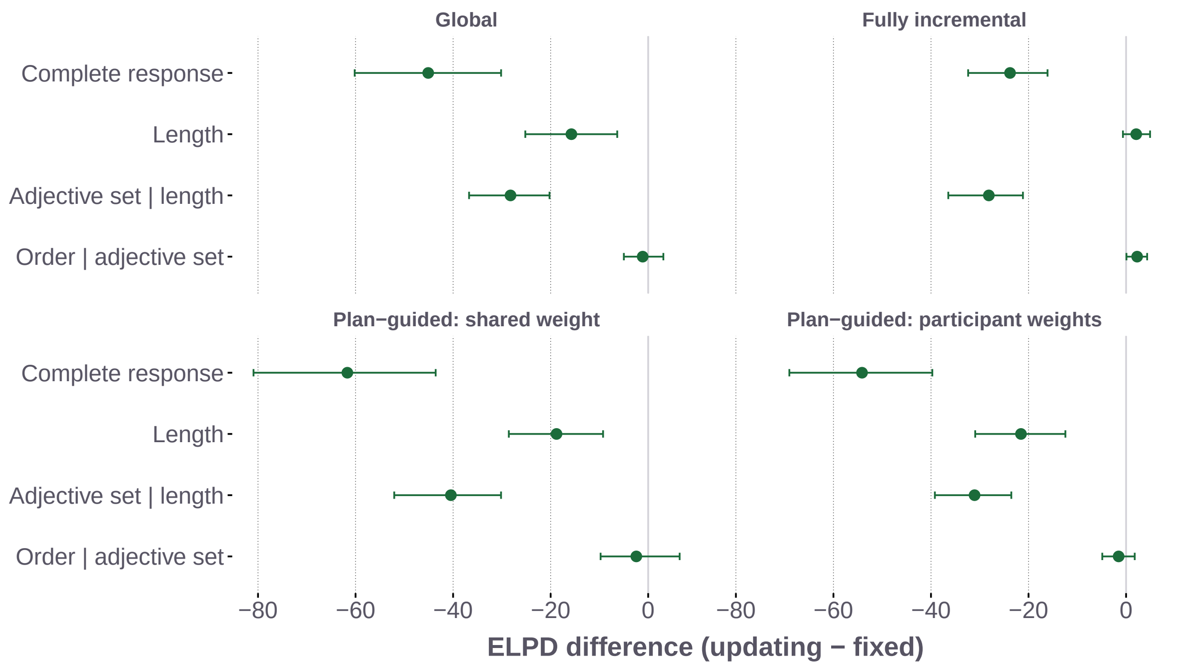

Figure 13: Additive decomposition of updating-minus-fixed response ELPD for each matched semantic pair. Length, adjective set conditional on length, and order conditional on the adjective set sum to the complete-response contrast. Points show paired score differences and intervals show participant-level Bayesian-bootstrap 95% credible intervals. The final panel uses participant-specific incremental-choice weights; the other plan-guided panel uses a shared weight.

Response residuals. We compared the two participant-weight plan-guided models using the posterior mean probability of each utterance in each combination of adjective family, referential context, and size discriminability. For each of the 270 response cells, the residual was the observed response proportion minus the mean predicted probability, expressed in percentage points. Positive residuals indicate underprediction and negative residuals indicate overprediction. These descriptive residuals use the full-data posterior; the ELPD decomposition above evaluates held-out responses. Across cells, mean absolute residuals were 3.04 percentage points for context-fixed semantics and 3.08 for updating; root mean squared residuals were 6.03 and 6.07, respectively. Table 8 shows the six cells with the largest absolute between-model changes in absolute residual error, retaining changes that favoured either model. All six occurred in size-sufficient displays. Updating increased the overprediction of size–colour–form utterances and the underprediction of size–form utterances in the size–form family, while reducing errors for size-only responses in both size-containing families under high discriminability. The held-out score partition likewise located the largest net disadvantage of updating in the size–form family, as reported above.

C.4 Posterior-Predictive Adequacy

The condition-wise response distributions appear in Figure 9. Across the 270 cells formed by the fifteen completed utterances, three adjective families, three referential contexts, and two size-discriminability conditions, the poste-

Table 8: Largest changes in response residuals between the participant-weight plan-guided semantic models. All rows concern size-sufficient displays. Observed proportions are percentages; residuals are observed minus predicted proportions in percentage points. S, C, and F denote size, colour, and form, with surface order preserved.

Adjective family	Discriminability	Utterance	Observed	Fixed residual	Updating residual
Size–form	High	S–C–F	21.1	−6.2	−10.1
Size–colour	High	S	27.2	2.7	0.9
Size–form	High	S	25.5	−2.3	−0.6
Size–form	Low	S–C–F	27.9	−4.1	−5.7
Size–form	High	S–F	21.5	12.9	14.4
Size–form	Low	S–F	19.8	8.8	9.8

rior predictions correlated with the observed proportions at $r = .909$, with a mean absolute error of 3.0 percentage points and a root mean squared error of 6.0 percentage points. The corresponding 95% posterior-predictive intervals contained 58.5% of the observed cell proportions, a descriptive check of fit across the dependent response cells.

D Model-Development Checks

Model development used the present data to select the form semantics, stable-order feature, and task terms. The matched comparisons hold these components constant across production architectures and semantic regimes.

The reported models fix form reliability at .50 and include a coefficient for form use, the GPT-2 order residual, sufficient-single-adjective support, a three-adjective penalty, and size-discriminability terms. Exploratory variants used informative form meanings tied to colour reliability, alternative priors on the form-use coefficient, an additional reliability-based order term, or no stable-order score. A reduced specification also removed the task terms as a bundle. These variants were evaluated before the semantic implementation was corrected and were not refitted. They informed model development; the predictive comparisons reported here use the common specification in Appendix B.

E Participant Variation and Predictive Contributions

The joint model estimates pragmatic optimality, incremental-choice weight, and stable-order weight for each participant. Table 7 reports its predictive gain relative to the otherwise matched shared-κ model under each semantic regime. In the best-predicting context-fixed model, the population location of the participant-specific incremental-choice weights was $\mu_\kappa = .472$, with a 95% credible interval of $[.337, .618]$. The participant distribution was broad, with an estimated logit-scale standard deviation of 2.720, 95% credible interval $[2.277, 3.207]$.

Participants whose posterior mean κ_i lay farther from the shared $\kappa = .446$ gained more participant-level ELPD from the hierarchy. The observed correlation was $r = .618$, and the participant-level Bayesian-bootstrap mean was .639, with a 95% credible interval of $[.525, .763]$ and posterior probability above zero greater than .999. At the response level, the hierarchy gained 229.372 ELPD on two-adjective utterances, 95% credible interval $[163.012, 319.334]$, and 183.796 ELPD on three-adjective utterances, 95% credible interval $[100.757, 304.826]$. Its contribution to one-adjective utterances was -48.426 ELPD (95% credible interval $[-70.371, -28.666]$).

Posterior-predictive contrasts between participants in the lower and upper tails of each parameter describe the associated differences in utterance choice. Higher α_i concentrates probability on a sufficient single adjective, especially S in size-sufficient displays and F in form-sufficient displays. Higher κ_i mainly redistributes probability among form-involving responses whose continuation sets differ after the first adjective. Higher β_i shifts probability from single-adjective responses towards longer responses that follow the stable order score. In the context-fixed joint model, the mean participant-level posterior correlations were .010 for α_i with κ_i and .025 for α_i with β_i. The posterior participant-level association between κ_i and β_i was positive, with mean $r = .162$ and 95% credible interval $[.073, .249]$. These correlations and tail contrasts describe the associations between participant parameters and response probabilities within the joint fit.

Three-adjective utterances accounted for 1,633 of the 9,100 modelled responses, or 17.9%. Within the fitted model, these responses balance accumulated referential value and salience against a shared three-adjective cost, with α_i, κ_i, and β_i determining how that balance is expressed for each participant. These quantities jointly determine the predicted frequency and ordering of three-adjective utterances.

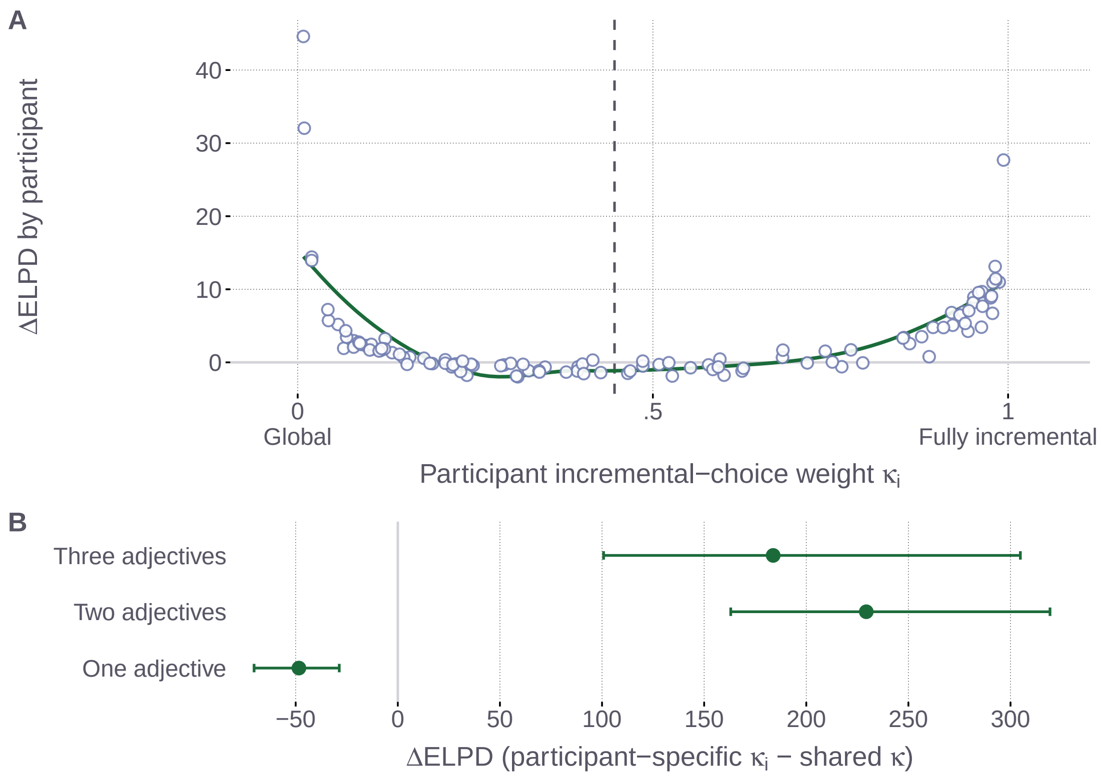

Figure 14: Where participant-specific incremental-choice weights improve prediction. Panel A relates each participant's leave-one-out gain from the κ_i hierarchy to their posterior mean κ_i; the dashed line marks the shared-κ estimate and the curve is a descriptive smooth. Panel B decomposes the total ELPD gain by observed response length. Intervals in Panel B are 95% credible intervals from the participant-level Bayesian bootstrap. Both panels use the best-predicting plan-guided model and its otherwise matched shared-κ model to localise the held-out model contrast.

F Local Normalisation and Model Endpoints

Local adjective competition. For the utilities defined in Equation (2), taking the logarithm of the product of locally normalised choices gives the accumulated utility minus the local-normalisation term. Suppressing the shared referent and scene arguments,

$$\begin{aligned}\log \prod_t \frac{\exp U_t(w_t \mid u_{<t})}{\sum_{a \in \mathcal{A}(u_{<t})} \exp U_t(a \mid u_{<t})} &= \sum_t U_t(w_t \mid u_{<t}) - \sum_t \log \sum_{a \in \mathcal{A}(u_{<t})} \exp U_t(a \mid u_{<t}) \\ &= U^{\text{complete}}(u) - \mathcal{N}^{\text{local}}(u).\end{aligned} \tag{18}$$

Relation between the endpoints. At common parameter values, plan-guided production with $\kappa = 0$ exactly recovers global production, and with $\kappa = 1$ exactly recovers fully incremental production. The stable-order score, task link, and utterance space are identical across the three architectures. Numerical checks found agreement in category probabilities and log probabilities within a tolerance of 10^{-10}.

G Sequential Context Updating for Size

For the hierarchical semantic path, let C_{j-1} be the context state after the syntactically inner modifiers have composed. The semantic regime determines the comparison class used for size:

$$[\![\mathrm{S}]\!]_{\text{fix}}(o) = [\![\mathrm{S}]\!]_{\theta_\delta(\mathcal{C})}(o), \tag{19a}$$

$$[\![\mathrm{S}]\!]_{\text{upd},C_{j-1}}(o) = [\![\mathrm{S}]\!]_{\theta_\delta(C_{j-1})}(o). \tag{19b}$$

where θ_δ derives a graded size threshold from a comparison class and $\mathcal{C}$ is the original scene, represented by the uniform prior P_0. The context-updating listener recomputes the size threshold from the hierarchically prior state; the context-fixed listener anchors it to the scene prior. At each surface position, the production calculation evaluates the current prefix using this noun-outward interpretation. Each prefix is interpreted from the original object prior, with each modifier applied once.

H Simulation Parameter Sweeps

The original two-order simulation compares a global rule based on completed-utterance utility with an incremental rule that includes continuation-versus-stop competition. We report its scene construction, definitions, and parameter sweeps first. Appendix H.1 then introduces the production specification on the same scenes, as in the controlled comparison in Section 6.

Each random display contains between 2 and 30 objects whose sizes are drawn from a normal distribution centred at 15.5 and truncated to $[1, 30]$. The largest object is designated as the target and assigned blue; each non-target object is independently assigned blue or non-blue with equal probability. Three levels of contextual size spread vary how distinctly the largest target differs from its competitors. The semantic grid varies colour reliability, size-threshold selectivity, and perceptual blur, which control the strength of colour-based restriction, how selectively *big* applies, and how sharply size differences are represented. In the simulation notation, k corresponds to threshold selectivity δ and w_f to perceptual blur w in Equations (9) and (10).

Figures 16 to 18 show how the three fixed semantic parameters affect communicative success across the two speaker architectures. Each figure separates the semantic regimes and averages equally over the other parameter settings. For the two-order simulation, write $q(v, o) = L(o \mid v, X)^\alpha$, with $\alpha = 1$ and equal utterance costs. For distinct adjectives a, b, denote the original global and incremental rules by S_G and S_I:

$$S_G(ab \mid o, X) = \frac{q(ab, o)}{q(ab, o) + q(ba, o)}, \tag{20}$$

$$R_I(ab \mid o, X) = \frac{q(a, o)}{q(a, o) + q(b, o)} \frac{q(ab, o)}{q(ab, o) + q(a, o)}, \tag{21}$$

$$S_I(ab \mid o, X) = \frac{R_I(ab \mid o, X)}{R_I(ab \mid o, X) + R_I(ba \mid o, X)}. \tag{22}$$

The second factor of R_I is the continuation-versus-stop choice, and the final normalisation conditions on producing a completed two-adjective order. The original global rule supplies the complete-utterance evaluation used in the first row of Panel A in the main comparison. The production global model accumulates utility across adjective positions; its incremental endpoint assigns utterance length through the completed-response distribution. The pragmatic listener uses a uniform referent prior, giving $L^{\text{prag}}(o \mid ab, X) = S(ab \mid o, X)/\sum_r S(ab \mid r, X)$. The implementation clips very small intermediate probabilities at 10^{-20}.

The reproducible sweep uses seed 20260906 and samples 1,000 independent scenes for each context-size and size-spread combination. Each scene is reused across semantic parameters, semantic regimes, and speakers, yielding 24,000 independent scenes and 12,000,000 model evaluations. Monte Carlo standard errors are calculated after averaging parameter settings within scenes; repeated evaluations of one scene do not count as independent samples. The computations use CUDA and 64-bit arithmetic.

Figure 15 shows how the size-first advantage changes across the broader random contexts, in which colour generally restricts the comparison class without uniquely identifying the target. Wider size distributions generally make the largest target more distinguishable from its competitors; clustered sizes reduce this distinction.

The original global rule with context-fixed semantics provides a symmetry baseline: both orders have the same fixed intersective meaning and equal cost, so they yield the same target probability. Sequential context updating breaks this symmetry because *big blue* interprets size relative to the comparison class established by colour, producing a size-first advantage under global production. The advantage is strongest in small contexts with widely separated sizes, where restricting the comparison class has the clearest consequences for which objects count as big, and weakens as contexts grow or sizes cluster.

The original incremental rule adds a contribution from local informativity along the surface sequence. With closely clustered sizes, this rule yields a colour-first advantage in pragmatic target recovery. As sizes spread apart, the colour-first advantage weakens. After averaging over the semantic parameter settings in Figure 15, the original incremental rule yields colour-first advantages in some contexts; the original global rule remains symmetric under fixed semantics and yields a size-first advantage under updating. At individual parameter settings, the original global rule with updating can also yield small negative estimated advantages. Under the original incremental rule,

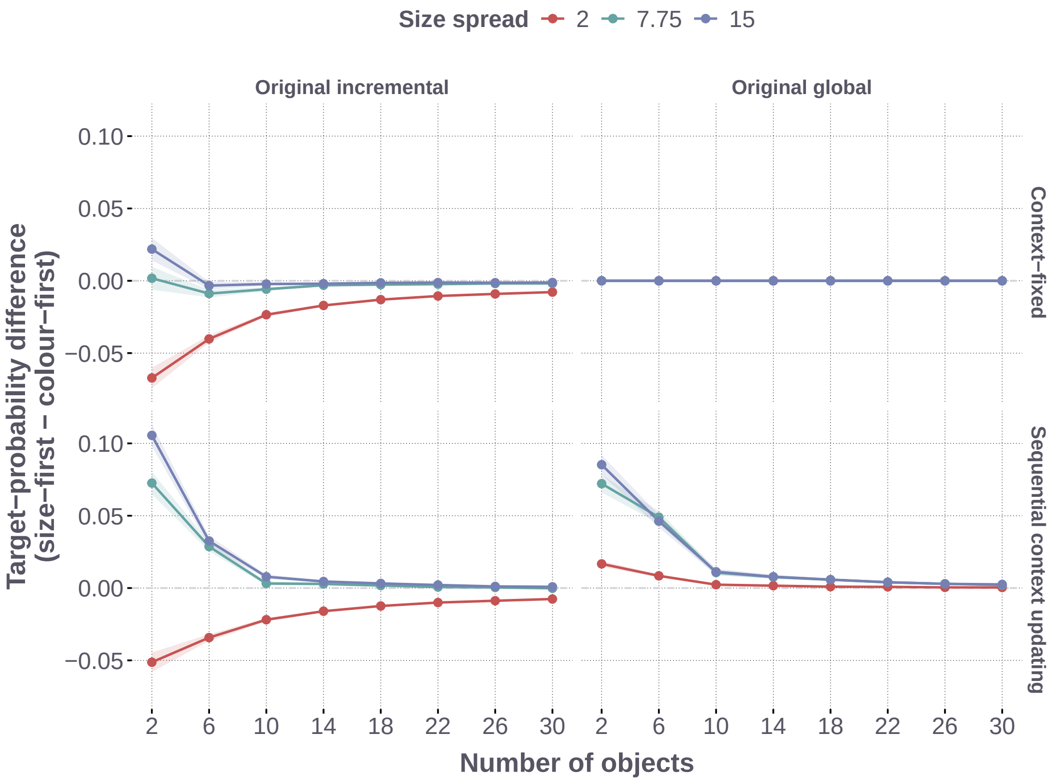

Figure 15: Size-first advantage Δ_{SF} across context size and contextual spread, shown for the original two-order global and incremental rules under context-fixed semantics and sequential context updating. Wider size distributions make size a more reliable identifying cue; clustered distributions make it less reliable. Each point averages over the semantic sensitivity settings; ribbons show ± 1.96 Monte Carlo standard errors calculated after averaging those settings within each sampled scene. The complete parameter grid appears in Table 9. Positive values indicate a higher model-assigned target probability for the size-first order.

sequential context updating increases the size-first target-recovery advantage when the colour-restricted comparison class changes size interpretation sufficiently strongly. The parameter sweeps in Figures 16 to 18 show that the same competition extends across colour reliability, size-threshold selectivity, and perceptual blur.

Table 9: Simulation parameter grid

Parameter	**Grid values**	**# levels**
Original two-order rule	Incremental, global	2
Semantic regime	Context-updating, context-fixed	2
n_{obj}	2, 6, 10, 14, 18, 22, 26, 30	8
σ_{spread}	2.0, 7.75, 15.0	3
Colour reliability ν	0.90, 0.92, 0.94, 0.96, 0.98	5
Threshold selectivity k	0.10, 0.30, 0.50, 0.70, 0.90	5
Perceptual blur w_f	0.1, 0.2, 0.3, 0.5, 0.8	5
Scenes per context-size/spread pair	1,000	
Total model evaluations	**12,000,000**	

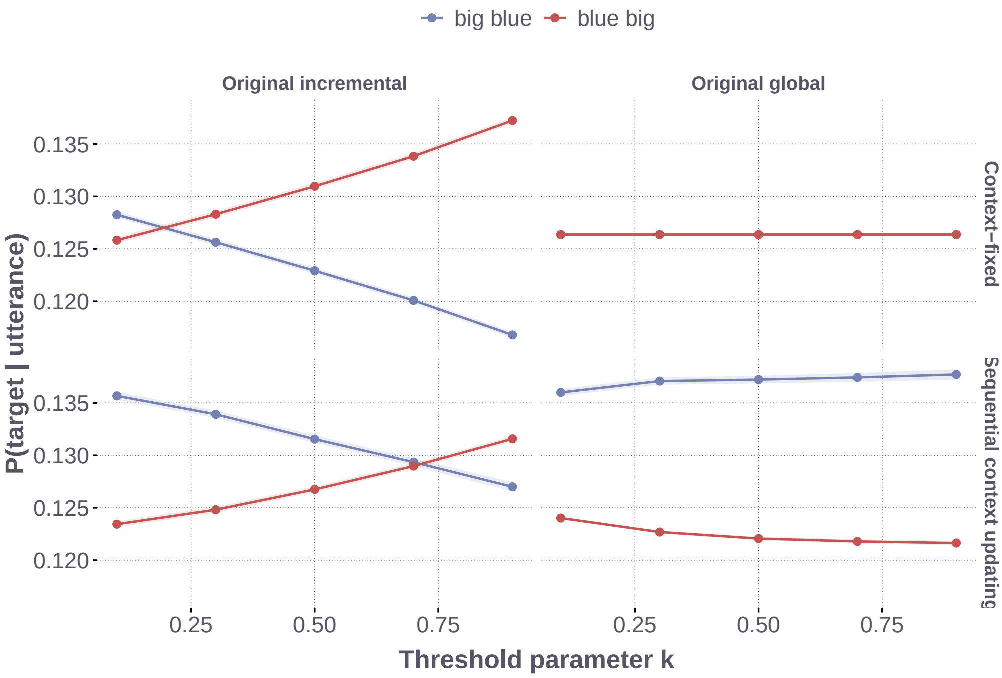

Figure 16: Effect of the threshold parameter k on pragmatic target recovery under the original two-order rules, separated by speaker and semantic regime. Curves average over the remaining settings; ribbons show ± 1.96 scene-level Monte Carlo standard errors.

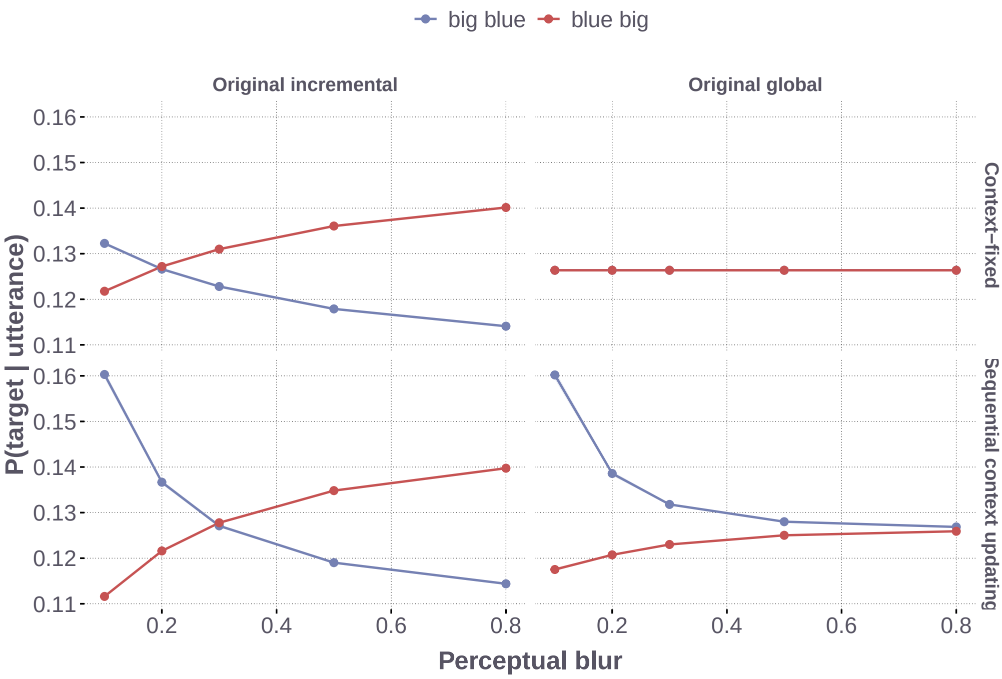

Figure 17: Effect of perceptual blur w_f on pragmatic target recovery under the original two-order rules, separated by speaker and semantic regime. Curves average over the remaining settings; ribbons show ± 1.96 scene-level Monte Carlo standard errors.

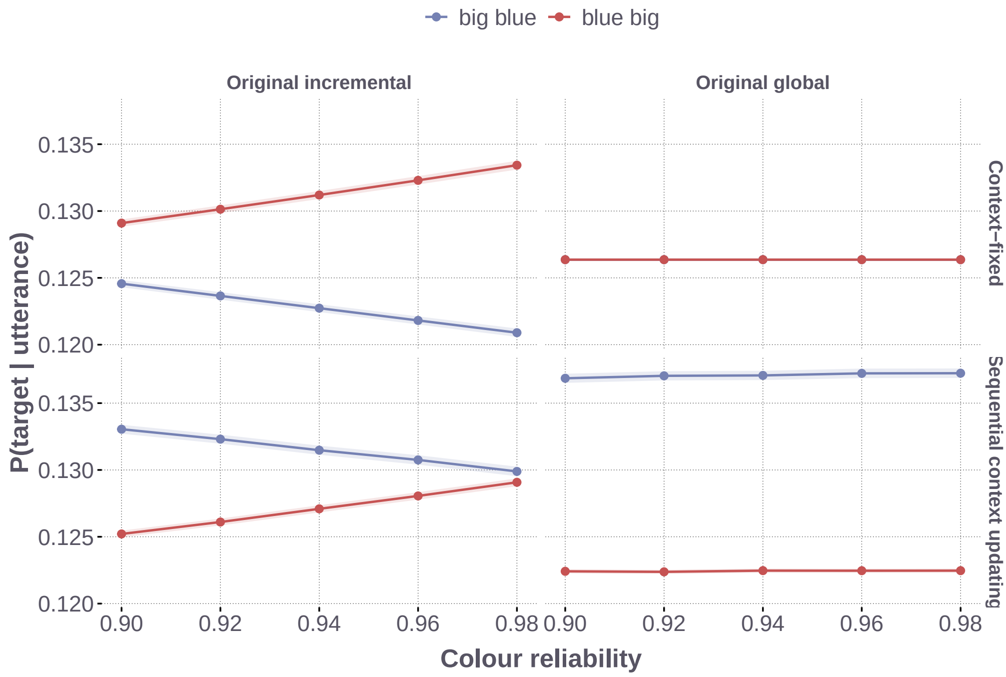

Figure 18: Effect of colour reliability ν on pragmatic target recovery under the original two-order rules, separated by speaker and semantic regime. Curves average over the remaining settings; ribbons show ± 1.96 scene-level Monte Carlo standard errors.

H.1 Controlled Changes on the Same Random Scenes

Complete-utterance evaluation, labelled terminal evaluation below, uses a different utility from incremental evaluation, labelled prefix evaluation:

$$U_{\text{term}}(u, o, X) = \alpha \log L(o \mid u, X), \tag{23}$$

$$U_{\text{prefix}}(u, o, X) = \alpha \sum_{t=1}^{|u|} \log L(o \mid u_{1:t}, X). \tag{24}$$

Both scores are globally normalised over completed utterances. The prefix score with $\kappa = 0$ is the global model used in the main text; $\kappa = .5$ and $\kappa = 1$ give the plan-guided and fully incremental simulation models. The terminal score supplies a control for the additional contribution of intermediate adjective information.

The controlled comparisons reuse all 24,000 scenes from the original sweep and the target-recovery measure in Equation (8). Each adjective retains the same extension when the intended referent changes; the referent prior is uniform. The response spaces contain two orders, four size–colour responses including the single adjectives, or the fifteen production responses. For expanded spaces, the pragmatic listener uses each order's probability under the full speaker distribution, without first conditioning on the two orders. The comparisons include terminal evaluation and prefix evaluation at $\kappa = 0, .5, 1$, using the original 125 semantic settings or the production constants. Prefix scores and local candidate sets are recomputed for each utterance space. They retain the production convention of no separate STOP action and final normalisation over completed paths. The full fifteen-response implementation agrees with the production forward calculation with task, salience, and order terms set to zero and final Laplace smoothing omitted. This isolates the computational components without assigning fitted participant effects or experimental task labels to random scenes.

The main comparison holds pragmatic optimality at $\alpha = 1$ and excludes stable-order expectations, salience, task-specific response terms, and final Laplace smoothing. Form reliability is fixed at .50. Panel A of the main figure separates the effects of evaluating information at each adjective position and expanding the utterance space, while holding global choice fixed. Panel B compares global, plan-guided, and fully incremental production using $\kappa = 0, .5, 1$, respectively, at the production semantic constants. The production constants are colour reliability .59, threshold selectivity .50, and perceptual blur .6856. Table 10 reports the mean advantages in both panels of Figure 10. The prefix-evaluation rows specify the architecture through κ; terminal evaluation uses global choice.

For the fifteen-response global model at the production constants, the paired updating-minus-fixed difference was 0.0037 percentage points, with a Monte Carlo interval of $[0.0007, 0.0067]$.

Table 10: Mean size-first pragmatic target-recovery advantage in percentage points on the same 24,000 random scenes.

Evaluation	Responses	Semantic settings	Context-fixed	Updating
Terminal	2	Parameter grid	0.000	1.141
Prefix ($\kappa = 0$)	2	Parameter grid	-2.718	-1.838
Prefix ($\kappa = 0$)	4	Parameter grid	2.257	2.772
Prefix ($\kappa = 0$)	15	Parameter grid	2.589	3.104
Prefix ($\kappa = 0$)	15	Production constants	1.891	1.894
Prefix ($\kappa = .5$)	15	Production constants	1.455	1.457
Prefix ($\kappa = 1$)	15	Production constants	0.960	0.962

The main figure uses the production semantic implementation throughout. As a separate implementation control, substituting its lexical calculation into the original two-order global rule reduces the mean updating advantage from 1.465 to 1.142 percentage points across the original grid. The terminal control additionally retains the production listener log-probability floor, giving the 1.141-point advantage in Table 10. The implementations differ in probability floors and numerical normalisation around weighted quantiles; the control measures their combined impact. The original curves are retained in Figure 15.

Figure 19 reports the fifteen-response comparison across evaluation rules and local-normalisation weights. The final panel adds the same stable-order coefficient, $\exp(.962357)$, to every model at the production semantic constants. This representative setting comes from the fixed model's population log-weight location; it does not integrate posterior or participant uncertainty. At $\kappa = 0$, adding this term gives mean advantages of 2.154 and 2.157 percentage points for fixed and updating semantics, respectively. The positive advantages are already present without this term. All other salience and task terms remain zero, and $\alpha = 1$ throughout.

For every contrast, semantic parameter settings are averaged within each scene before calculating Monte Carlo uncertainty. The overall mean gives equal weight to the 24 context-size and size-spread groups. These averages summarise the specified parameter grid; individual contexts can favour different orders. Reproduction checks recovered the stored two-order outputs, and the controlled probabilities agreed with the corresponding production calculation to numerical precision.

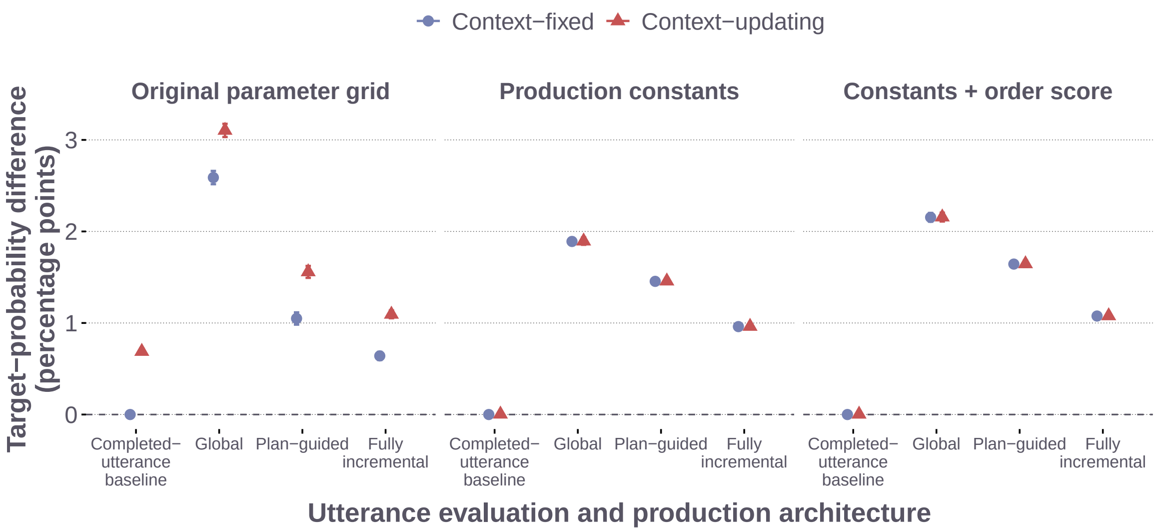

Figure 19: Controlled fifteen-response comparison across evaluation rules, semantic parameters, and a shared stable-order score. The completed-utterance baseline uses only the final literal-listener target probability. The production models accumulate prefix scores: $\kappa = 0$ gives global production, $\kappa = .5$ plan-guided production, and $\kappa = 1$ fully incremental production. Panels use the original semantic grid, the production semantic constants, and those constants with the stable-order score added. Points and intervals show mean target-probability differences and ± 1.96 scene-level Monte Carlo standard errors. All comparisons use the same random scenes; no participant posterior uncertainty is represented.

H.2 Supplementary Predictions on Experimental Displays

The supplementary simulation uses the 3,034 size–colour trials and all eight fitted production models. For each model, 50 evenly spaced retained draws from each of its four chains give 200 posterior parameter draws. The simulation preserves the fifteen named responses, fitted semantic constants, participant parameters, stable-order score, salience calculation, and task link. For each trial and posterior draw, each of the six objects is evaluated as the intended referent while the adjective extensions and observed task covariates remain fixed. Thus changing the candidate referent does not change which colour or form the uttered words denote. Because the production responses were coded by adjective type, fixing these named adjective extensions across candidate referents is an additional assumption of the listener calculation. This is a model-based extension to counterfactual referents within the observed task context.

Let $S_{it}^{(d)}(u \mid o, X)$ be the resulting complete-response probability at posterior draw d. We evaluate the speaker's size-first probability conditional on the size–colour adjective set as

$$\frac{S_{it}^{(d)}(\mathrm{SC} \mid o^*, X)}{S_{it}^{(d)}(\mathrm{SC} \mid o^*, X) + S_{it}^{(d)}(\mathrm{CS} \mid o^*, X)}. \tag{25}$$

Separately, the pragmatic listener normalises the full-response speaker probabilities over the six referents using a uniform prior. The listener therefore retains the fitted fifteen-response normalisation when evaluating each order. We average each quantity over the original size–colour trials within posterior draw, then report the posterior mean and central 95% interval. Replaying the original target reproduces its stored likelihood within 10^{-8} for every model.

Figure 20 reports the resulting speaker and listener predictions. In the participant-weight plan-guided models, the mean size-first target-probability advantage was .0062 under fixed semantics, 95% posterior interval $[.0047, .0077]$, and .0058 under updating, $[.0045, .0072]$. The corresponding mean probabilities of size-first conditional on the size–colour adjective set were .967 and .966. Both fitted regimes therefore generate a size-first advantage under this counterfactual listener calculation. In the response-score decomposition, the predictive difference between these models was concentrated in adjective content and length. The two calculations describe how the models differ in predicting the choice of a description while both assigning a listener advantage to size-first order.

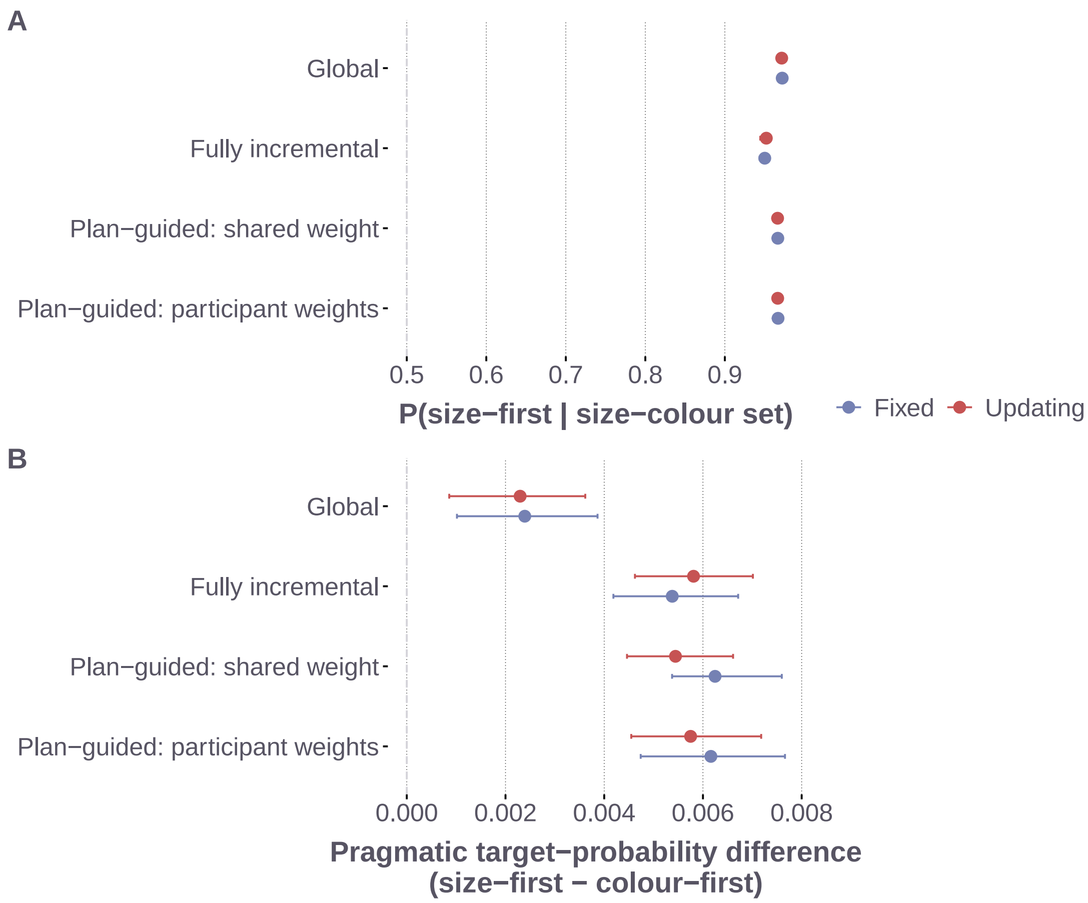

Figure 20: Supplementary predictions from the fitted production models on the size–colour displays. Panel A shows mean speaker probability of size-first conditional on selecting the size–colour adjective set. Panel B shows the difference in pragmatic target probability between size-first and colour-first. Both panels preserve the fitted fifteen-response distribution; the listener in Panel B uses a uniform prior over six referents. Intervals are central 95% posterior intervals over trial-averaged quantities using 200 posterior parameter draws per model.